MALMÖ UNIVERSITY

Speculative Exploration on Empathetic Design Solutions for Animal IDs

Designing with animals and humans,
for animals and humans

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-- Note --

Going forward in this thesis, the author will refer to herself as ***we*** instead of ***I*** to include her furry companion, Ori, who inspired the exploration.

ABSTRACT

What began as an exploration into alternative, non-invasive pet identification methods evolved into a reflective journey that challenged and reshaped our understanding of animals in society and the concept of identification. Alongside traditional research methods, we embraced a speculative approach, imagining and critically reflecting on bits of a future world where pets have IDs. Through interconnected phases of exploration, iteration, and reflection, we explored the implications of pet identification and designed with animals and humans, for animals and humans.

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1 INTRODUCTION

In contemporary society, the status of animals, particularly pets, has undergone a profound transformation, shifting from mere companions to cherished family members with recognized individual rights. According to the European Union Council, in 2023, nearly half of European Union citizens declared *owning** companion animals (European Union, 2024, p. 5). The welfare of companion animals is an important subject, and there is widespread opinion that they should be better protected than now.

In Europe, the first official documents to consider and enact national animal welfare laws are dated as early as the 1800's. Paving the way is the United Kingdom's *Act to Prevent the Cruel and Improper Treatment of Cattle* in 1822 (Politics.co.uk., 2016; Great Britain Parliament, 1822), followed by other countries, including Sweden's *Criminal Law* in 1857 (Striwing, 2002). Although primarily focused on addressing public mistreatment of animals as a violation of decency rather than directly addressing the animals' suffering, these laws nonetheless served as foundation stones for our current legislations.

In 1997, a significant milestone was reached when animals were officially recognized as *sentient beings* in the EU legislation (European Union, 1997, p. 110), a term that, since then, has been reinforced in other official documents (European Union, 2007, p. 51). This legal recognition marked a profound shift in the status of animals, reflecting a broader cultural evolution toward greater empathy and consideration for non-human beings, challenging traditional notions of ownership and control.

In this context, it is important to reconsider the language we use to describe our relationships with animals. The term *pet owner* is commonly and legally used to identify individuals listed on the pet's identification document (European Union, 2016, p. 5) and, in most cases, living with and caring for non-human companions. However, this term implies ownership and control, which do not fully capture the evolution and complexity of the human-animal relationship. Instead, alternative terms such as *pet guardian*, *companion caregiver,* or *pet caretaker* more accurately represent the reciprocal nature of this relationship, highlighting the shared responsibility for the welfare of animals.

Increasingly, pets are being recognized not as property to be owned but as individuals deserving of respect, care, and autonomy. As our understanding of this relationship evolves, it becomes essential to reconsider the language we use to describe it, adopting a more respectful and inclusive tone. By acknowledging the importance of care in our interactions with animals, we not only honor our mutually beneficial bonds but also reaffirm our commitment to

building a more compassionate and harmonious world for all beings, both human and non-human alike.

1.1 Research Questions

This project is an exploration towards empathetic design solutions, within a speculative [hopefully near] future in which animals have Pet Identification Cards [Pet IDs].

Going forward our focus lies with the following species, currently under the “pet animal” label in the EU legislation (European Union, 2016, Annex I):

1. Dogs (*Canis lupus familiaris*)
2. Cats (*Felis silvestris catus*)

Going forward the research questions are set as follows:

1. How has the evolving societal perception of pets influenced the design requirements for pet identification systems?
2. What are the primary needs and challenges faced by pet caretakers, and how can interaction design address these concerns?
3. How can collaborative design methodologies involve all stakeholders, humans or not, in the design process, and what interdisciplinary collaboration opportunities exist?
4. How can a pet identification solution balance data collection with considerations for usability, privacy, and ethical treatment of animals?
5. What are the potential long-term benefits and implications of widespread adoption of pet identification systems, and how do they impact stakeholders' lives?

2 BACKGROUND

2.1 Identification

In the Merriam-Webster.com dictionary, "**identification**" [*ī-ˌden-tə-fə-ˈkā-shən*] is defined as evidence of identity. The concept of identification, from this project's perspective, refers to the process and systems through which individuals, objects, or entities based on their unique characteristics, attributes, or properties are recognized, labeled, distinguished, and represented within various contexts.

Identification serves several various purposes across different domains, including establishing identity, ensuring security, enabling efficient operations, and facilitating personalized services across various contexts, ultimately contributing to safety, accountability, and effective management.

The need to recognize and differentiate has been intrinsic to human societies, from ancient symbols and seals (University College London, 2001) to modern biometrics (Biometrics Institute, n.d) and digital IDs (Finansiell ID-Teknik BID AB, 2024), identification as a system had a remarkable evolution throughout human history, reflecting the changing needs, priorities, and capabilities of societies. Learning from the past we can assume the systems will continue to adapt further on and designers play a crucial role in its journey.

In the tables below (Table 1 & Table 2), we mention the main types of identifiers in humans and animals, split into three main categories based on their localization: inside, on or outside the body.

2.1.1 Identification in humans

For humans, identification means to establish identity, while often involving the assignation of labels or categories to individuals or groups based on factors such as name, gender, age, nationality, or social status. But can labels lead to discrimination?

Location	Category	Time	Examples
Inside the body	Biometrics	Permanent	Fingerprints, Facial features, Iris Patterns, Voice, DNA
On the body	Branding	Permanent	Tattoos
	Wearables	Semi-Permanent	Jewelry, Tech

Outside the body	Wearables	Temporary	Jewelry, Tech
	Personal Identifiers	Permanent	Social Security number
	Documents	Temporary	Passport, ID Card

Table 1 *– Identification methods in humans*

2.1.2 Identification in non-humans: animals

The main roles of identification in animals are in healthcare and traceability of the animal, facilitating responsible *ownership*, disease prevention and control measures for movements into and around the European Union. (European Commission, 2022). From the main identification methods below (Table 2) we chose to focus on the ones located inside the body, as they offer a permanent option, without the risk of misplacing the unique identifying marker.

Location	**Category**	**Time**	**Examples**
Inside the body	Biometrics	Permanent	Fingerprints, Facial features, Iris Patterns, Nose Prints, Coat Patterns, Blood, DNA profiling, Pedigree (Genetics)
	Personal Identifiers	Permanent	Microchips / RFID Tags
On the body	Branding	Permanent	Tattoos, Hot Iron Marks, Ear Clipping, Tail Clipping
	Wearables	Semi-Permanent	Ear Tags, Leg Bands
Outside the body	Wearables	Temporary	Collar Tag, Harness
	Documents	Temporary	Passport, Health Card

Table 2 *– Identification methods in non-humans: animals*

2.1.3 Microchips in non-humans: animals

The most common way to identify pet animals is through a microchip, an electronic chip approximately the same size as a grain of rice (AVMA, 2024; Rizvi et al., 2016). It contains a unique identification number linked with the pet's caretaker's contact information in a national database. This identification method requires a quick medical procedure involving implanting the chip under the pet's skin, usually between the shoulder blades. (Dr. Lera @ Heron Lakes Animal Hospital, 2020)

In Europe, microchipping has become a standard practice and legal requirement (European Union, 2016; DG Health and Food Safety, 2022; Jordbruksverke, 2023) for pet identification. It provides an invasive but permanent and effective (Dr. Lera @ Heron Lakes Animal Hospital, 2020) method, making it a reliable and accurate way of identifying the pet in case they get lost or separated from the caretaker. Despite the health risks (Albrecht, 2010; Hamabe et al., 2022; Azoulay et al., 2023) and migration possibility (Lancaster et al., 2015; Azoulay et al., 2023) associated with this procedure, it is considered safe enough, and the benefits outweigh the potential risks. (AVMA, 2007; WSAVA, 2020)

2.1.4 Biometrics

Biometrics *[bī-ō-ˈme-triks]* is referring to the measurement and analysis of unique physical or behavioral characteristics especially as a means of verifying personal identity (Merriam-Webster Dictionary, 2024).

Biometrics have been often used for identification in security use cases, and you can label them into the following categories (Kaspersky, n.d; Biometrics Institute, n.d):

Category	**Explanation**	**Examples**
Biological	genetic and molecular level	DNA, Blood, Fluids
Morphological	structure of the body	Hands, Facial, Eyes
Behavioral	patterns unique to each person	Voice, Gait, Heartbeat

Table 3 *– Biometrics categories*

2.1.5 Biometrics in non-human: animals

Animal biometrics involves using animals' unique physical or behavioral characteristics for identification and authentication. Just like human biometrics, which includes fingerprints, iris scans, and facial recognition, animal biometrics relies on various distinct traits, specific to each individual animal. Depending on the species, these biometric traits can include both physical features such as fur patterns (Lahiri et al., 2011; Kumar & Singh, 2016), nose prints (Bae et al., 2021), facial recognition (Sagonas et al., 2017) as well as behavioral characteristics like gait, vocalizations, and other movement patterns. Unlike traditional methods such as tags or microchips, which can be lost, damaged, or removed, biometric traits are inherent to the animal and remain constant throughout its life.

Animal biometrics identification technology has advanced significantly in recent years thanks to developments in computer vision, machine learning, and image processing algorithms. These advancements have enabled the development of sophisticated biometric recognition systems capable of accurately identifying animals based on their unique traits, even in challenging environments or conditions.

However, like any technology, animal biometrics identification also raises ethical and privacy concerns, particularly regarding the collection and use of biometric data from animals. It's essential to consider the welfare and rights of animals when implementing biometric identification systems, ensuring that they are used responsibly and ethically.

In this thesis we will specifically use the theories that the unique biomarkers of animals, with a focus on nose prints and facial recognition in dogs and cats are an accurate method of identification. Research has shown that the intricate patterns and ridges on a dog's nose are highly distinctive (Coldea, 1994), allowing for accurate identification of individual animals (Choi et al., 2021). Moreover, unlike other non-invasive methods, such as tags or collars, that can be lost or removed, a dog's nose print remains constant throughout its life and is not subject to natural alteration. We can assume that dogs possess unique nose prints (Choi et al., 2021), which, like humans' fingerprints, can be used as a reliable biometric identifier.

2.2 ACI and HCI

Animal-Computer Interaction (ACI) is an interdisciplinary field that explores the interaction between animals and technology, with the aim of enhancing communication, understanding, and well-being for both animals and humans.

Although some designers argue that Animal-Computer Interaction (ACI) is an entirely different field than Human-Computer Interaction (HCI) (Zamansky et al., 2017), we would argue that this project is placed at their intersection, as it involves both human and animal stakeholders in the design process. While navigating this project, we will borrow methods from HCI and adapt them to our context, considering the differences in interspecies and communication barriers (Zamansky et al. 2017). For example, we must consider that non-human stakeholders do not have the same capabilities of abstraction and projection [imagination] as human stakeholders while making design decisions, such as choosing prototypes over sketches (French et al., 2017).

Aligning with Mancini's Manifesto (2011), this project's aim is to contribute to shaping the trajectory of animal identification technology development by focusing on three main points.

1. Firstly, we aim to empower animals in legal contexts by envisioning a Pet ID that redefines traditional identification methods and systems.

2. Secondly, we seek to strengthen the bond between humans and animals by exploring safe and effective ways to integrate and utilize the Pet ID within daily interactions.

3. Finally, we aspire to elevate the quality of life and longevity of animals by advocating for the widespread adoption of non-invasive identification methods that are both effective and hopefully, legally endorsed in the future.

By embracing a user-centric methodology grounded in empathy and respect for both animals and their caregivers, we hope to contribute to a more compassionate and inclusive future where technology serves to strengthen the bond between humans and animals, rather than perpetuate systems of dominance and control and inspire further research (Mancini, 2011) in the field of animal identification systems.

2.3 Ethical considerations

While designing with animals it is important to take a non-speciesist approach, respecting and caring for all participants equally (Mancini, 2011). The nature of this subject raises many ethical questions, some of which will be discussed in *Chapter 6*. Implementing a Pet ID means shifting the current power balance between humans and animals. Going forward with this research opens an invitation to reflect upon the identification systems we have today, for humans and animals alike.

3 STATE OF ART

3.1 Official Identification Documents

Official identification documents are government-issued credentials designed to facilitate travel and identification for individuals, humans, and animals. They are universally recognized by the law and are essential for crossing international borders and interacting with the governmental systems.

These documents typically hold essential information about the holder, such as name, date of birth, photograph, among other personal identifying details. In our current system, human citizens are each allocated a personal identification number tied to their national identity and found on official documents. Similarly, pets are assigned a unique identification number through their microchips. It is important to note that today, on official documents issued by the governments, the pet caretaker is still referred to as *pet owner*.

3.1.1 National Identity Card

A. For humans

National identity cards, or ID cards, are issued by governmental authorities to citizens or residents of a country and typically serve as the primary form of identification within a country. They are widely accepted as proof of identity for various purposes, including accessing governmental services, opening bank accounts, voting in elections, and conducting financial transactions.

The format and design of national ID cards (Figure 1) vary depending on the country issuing them, but they often resemble a small plastic card similar in size to a credit card. Many national ID cards incorporate security features such as holograms, watermarks, and embedded microchips to prevent counterfeiting and unauthorized use.



Figure 1 – Front of Swedish and Turkish National ID Cards

B. For pets

At this time, some existing equivalents of identity cards for pets could be the Health Notebooks (Figure 2a & 2b) or even the microchips themselves. The Health Notebook (ANSVSA, 2014) for pets is a physical record of health information about the pet, containing details such as vaccination history, medical treatments, allergies, and other health-related information. Unlike other official documents, the health notebook is a space for the pet's medical history rather than serving as a means of identification.

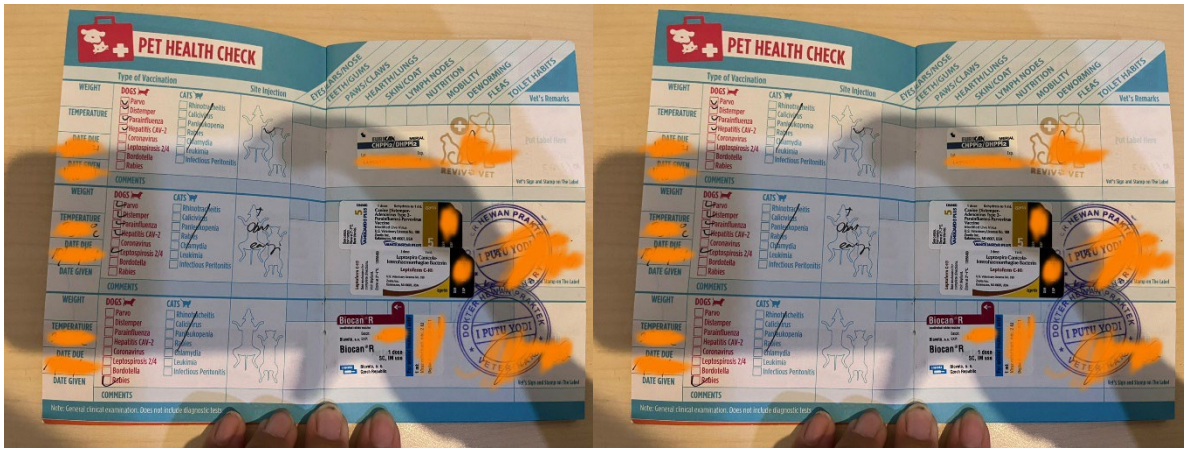


Figure 2a and Figure 2b – Indonisian Health Notebook

3.1.2 Passport

A. For humans

The human passport is an official document used globally for international travel and proving an individual's identity and citizenship. Issued by a national government, the passport contains personal information, including the personal number. It also includes a photograph and, in many modern passports, biometric data like fingerprints or iris scans to enhance security and prevent fraud (Orience, n.d.; U.S. Department of Homeland Security, 2023).

B. For pets

The Pet Passport is a national official pet identity document, issued by authorized veterinarians to caretakers and is mandatory for pets to cross international borders with their caretakers. Up to date, in Europe, the only animal species that can get passports are the pet animals recognized by the current legislation: dogs, cats and ferrets (European Union, 2016). These passports provide a recognized means of documenting an animal's health status and *ownership* information and ensuring compliance with relevant regulations.

Inside the European Union, member countries issue their bilingual pet passports following a standardized European small notebook template. The passports feature two languages: English and the country of origin's and have a unique identification number indicating the country (Figure 3a & 3b). They contain essential information about the pet and their caretaker. The standardized approach ensures that the information is consistent and reliable, regardless of the country issuing the passport.

The passport layout features the caretaker's information on the first page and a pet description on the second page, including a photo, name, breed, gender, birth date, color, and unique markings. The third page typically contains information about the pet's microchip, ensuring accurate identification and traceability (Figure 3c & 3d).

Besides the microchip documentation, among other mandatory information required are records of rabies vaccination and tapeworm treatment (Jordbruksverke, 2022).



Figure 3a and Figure 3b – Covers of EU Pet Passports [a.Romania & b.Denmark]

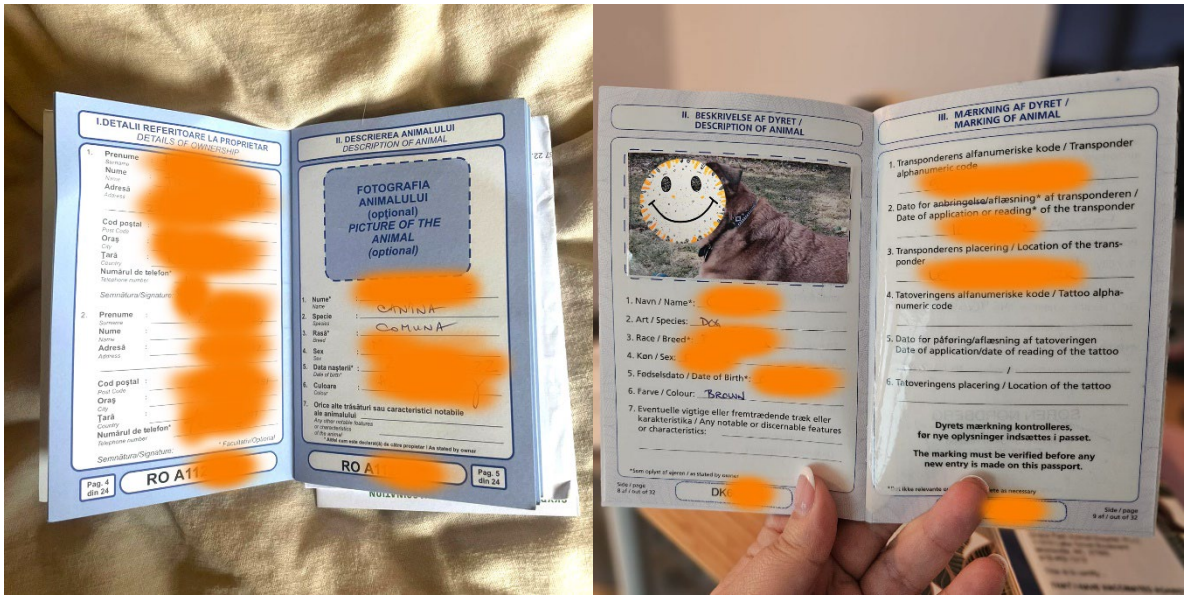


Figure 3c and Figure 3d *– First Sections of EU Pet Passports [c.Romania & d.Denmark] containing personal information about the caretaker, pet and its microchip*

3.2 Animal Biometric Identification Applications

Drawing inspiration from the success of human biometric recognition methodologies, such as fingerprint scanning and facial recognition, researchers and practitioners have increasingly turned their attention to adapting similar principles for animal identification purposes. These applications leverage artificial intelligence (AI) to identify animals based on their unique biometric traits, offering solutions for wildlife monitoring (Kumar et al., 2017), conservation efforts, and pet management.

3.2.1 Pet Now

PetNow is a pioneering mobile application, that has already implemented the biometric identification of dogs and cats with the help of AI technology. For dogs, this involves the distinct pattern of their nose print (Figure 4a), while for cats, it focuses on their facial features. It offers the possibility to register your pet into their database and create a profile. Additionally, it offers a feature where users can declare their pet missing and send alerts to nearby other users (Figure 4b).

Implementing biometrics and making them accessible through an app is a new approach to pet identification. Having this technology in their pocket encourages users to take steps to protect their pets as registering them in the database significantly improves their chances of recovery in case of loss.

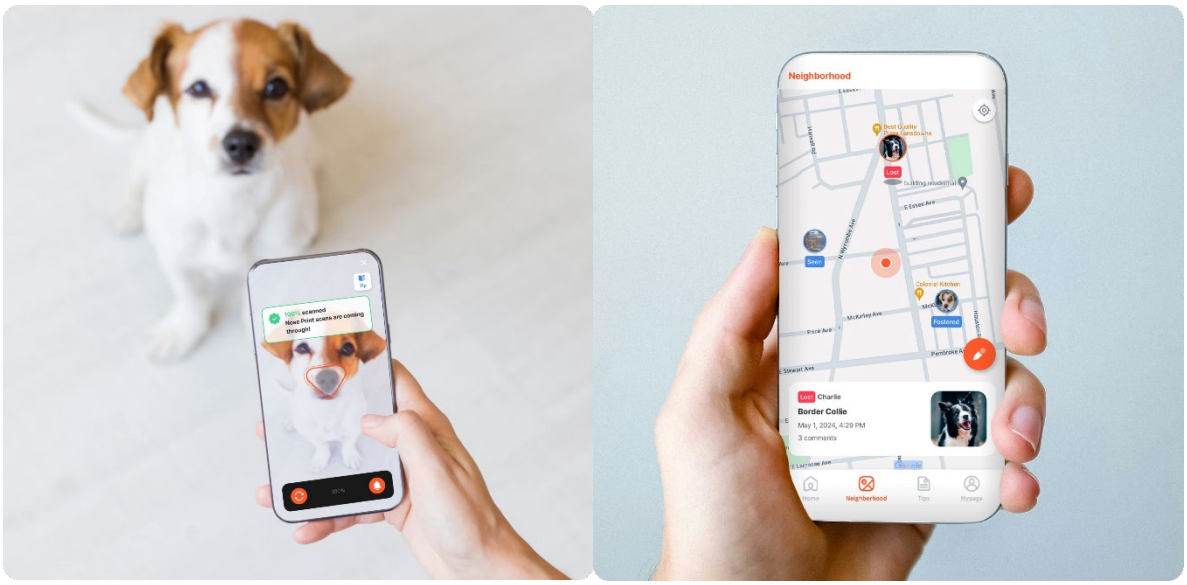


Figure 4a and Figure 4b – Images from the PetNow website https://petnow.io/en/

3.2.2 Snap Animal App

The Snap Animal App is a tool, still in development, aimed specifically at the agricultural sector, helping farmers to better manage their livestock. At its core stands an AI biometric animal identification system that allows for non-invasive identification of bovine animals, through face ID (Figure 5). It is also accessible (as easy to carry with you); the technology lives in your smartphone and grants you access to a database with your registered cattle animals.

Although this application targets livestock animals, it is an excellent example of how digital tools are being tailored to specific needs within the broader category of animal management, leveraging technology to improve efficiency, monitoring, and overall animal welfare.

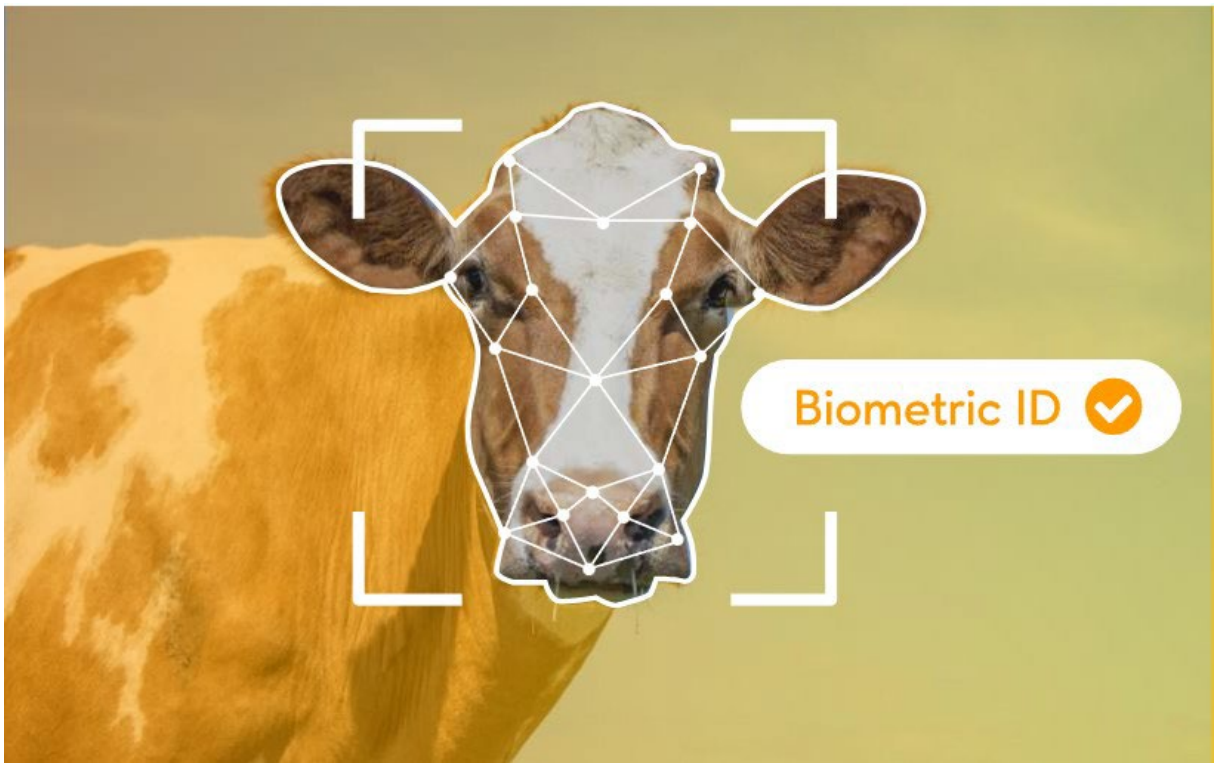


Figure 5 – Image from the Snap Animal website https://www.f4tlab.com/phaid

4 METHODS

4.1 Speculative Design

Speculative design is an approach within the field of design that explores and provokes discussions about possible futures, alternative realities, and emerging technologies through creative and imaginative artifacts, narratives, and scenarios (Dunne & Raby, 2013). Unlike traditional design practices that focus on solving present-day problems or addressing immediate needs, speculative design seeks to challenge assumptions, question existing norms, and stimulate critical thinking about the implications of technological, social, and cultural changes. It creates not only things, but ideas. It creates not only speculative futures, but also alternative presents that are shaped more responsibly and inclusively (Auger, 2013).

4.1.1 Design Fiction and Storytelling

The Design Fiction approach stands at the intersection of design, science, and science fiction (Bleecker, 2009). It is a journey that involves the creation of fictional narratives, characters, and artifacts, all of which serve as powerful tools for imagining alternative futures and reflecting on the implications of technological change.

Speculative scenarios are narratives that explore potential future uses of a product or service in various contexts and situations. Their primary goal is to make design ideas concrete, providing designers with insights into their potential impact on user's lives and how a product might be used in real-world scenarios (Hanington & Martin, 2021). By using speculative scenarios and the art of storytelling, we not only speculate about future scenarios and possibilities but also ground them into the present, acting as a bridge between the existing world and the future (Auger, 2013).

By presenting scenarios from the viewpoint of personas, designers can empathize with users' needs, motivations, and behaviors, thereby avoiding a narrow focus on technical requirements and instead designing culturally meaningful artifacts that enhance daily human activities (Cooper et al., 2007). Speculative scenarios are flexible in nature, allowing for multiple variations and interpretations. They are typically written as stories with minimal visuals, emphasizing the capabilities and benefits of the technology rather than its technical details due to its uncertain and fluid nature. In an age where technology evolves rapidly, so do the requirements. (Carroll, 1999).

4.1.2. Speculative Enactments and Role-playing

Role-playing in interaction design involves designers immersing themselves in the role of users and acting out scenarios to gain insights into user experiences and challenges. This method fosters empathy and highlights design opportunities by allowing designers to experience firsthand the routines and behaviors of users in realistic scenarios (Hanington & Martin, 2021).

The setup for role-playing is typically straightforward, often requiring nothing more than the people in the room. However, if more sophisticated environmental props are necessary, simulation exercises or bodystorming methods may be more appropriate (Buchenau & Suri, 2000). During role-playing, participants act out user scenarios guided by descriptions of situations, suggested actions, tasks, or goals. Improvisation is encouraged to approximate real-life situations. Role-playing is particularly valuable when direct observation of users is not feasible or ethical, such as in sensitive situations or when access to users is restricted.

By inviting participants to enact speculative scenarios, we create opportunities for co-creation, dialogue, and reflection therefore fostering a deeper understanding of the possibilities and implications of alternative futures. Speculative enactments encourage participants to consider different perspectives, challenge assumptions, and envision new possibilities for society, technology, and culture. (Elsden et al., 2017).

4.2 ANT

Actor-Network Theory (ANT) offers a distinctive lens through which to examine the complexities of social phenomena. At its' core is the idea that both human and non-human entities, or *actants* (Latour, 1996), are active and participants in shaping social reality and are seen as equal in terms of their capacity to act and exert influence within networks of associations. This approach challenges traditional heterogeneous perspectives that prioritize human agency over non-human agency.

ANT, as conceptualized by Latour, underscores the dynamic nature of social relations. It highlights the process of translation, an alignment of interests, meanings, and values between different actors within a network. This process is not static, but rather a continuous negotiation and reconfiguration of alliances and associations.

ANT has been used to study diverse phenomena, from developing and implementing new technologies (Davey & Tatnall, 2012) to organizing and governing social movements (Svein, 2013). By providing a framework for understanding the complex interplay between human

and non-human actors within networks, we aim to gain a new perspective in our exploration of animal Pet ID.

4.3 Literature Review

Literature review is an important research method in design projects that involves studying previous research on your topic of interest and including the essence. While the research may include a diverse range of references such as books, articles, websites, and documented projects, it is essential to choose credible sources (Hanington & Martin, 2021). Organizing the material by research categories is often useful. This method allows the researcher to adapt their methods accordingly.

4.4 User Interviews

User interviews are a research method used to gather data directly from a potential user by having a dialogue or even what could be called a “conversation with a purpose”. (Kahn & Cannell, 1957).

Unstructured interviews facilitate an in-depth exploration of user perspectives, motivations and experiences while offering flexibility for the participants to express themselves freely and provide nuanced insights. The key characteristic that distinguishes these types of interviews from the rest is their open-ended nature (Rogers, Sharp & Preece, 2019), having no expectations about the format or content of the answers thus allowing participants-centered discussions that adapt and possibly uncover unexpected new ideas and directions. Although they are labeled as unstructured, they still possess a skeleton that includes the main topics of interest.

Semi-structured interviews are a valuable qualitative research method utilized in interaction design to gather detailed insights from participants while maintaining a degree of flexibility and structure (Hanington & Martin, 2021). Guided by a set of core questions or topics, these interviews offer a balanced approach that allows interviewers to explore specific areas of interest while remaining open to unexpected insights. Participants are encouraged to express themselves freely, with interviewers probing for deeper insights and asking follow-up questions to enrich the conversation.

In this project we used user interviews to gather data directly from pet caretakers, allowing us to understand their needs, experiences, and challenges related to their pet. This method

helps validate our assumptions, build empathy, and gain deep insights into users' thoughts and feelings.

4.5 Synthesis & Affinity Maps Diagramming

An affinity map serves as a visualization tool used to organize and cluster observations and insights obtained from research activities, enabling designers to stay rooted in data throughout the design process (Hanington & Martin, 2021).

Gathering a lot of raw data early in the research process can become overwhelming. Affinity diagramming addresses this by giving designers a method to capture and sort the information based on affinity or similarity, form clusters and ultimately discover themes and patterns (Holtzblatt & Beyer, 1998). Analyzing the data we gathered led to valuable insights that guided the design process.

4.6 Sketching

Sketching is a fundamental tool for ideation and communication throughout the design process. It is seen as a means of externalizing and refining ideas, facilitating collaboration among designers, stakeholders, and users. Sketching serves to explore design possibilities rapidly and fluidly, allowing designers to generate and iterate concepts quickly (Buxton, 2010). By creating low-fidelity sketches, designers can convey their ideas without getting stuck in details, enabling them to focus on the broader conceptual aspects of interaction design.

While the roles of sketches and prototypes are complementary, they are not the same and should be used at different stages of the process (Rojas, 2023). Sketches are primarily used in the exploratory stages of design. They help propose, refine, communicate, and critique ideas in a tangible format without significant investment. Sketches also provide a shared language through which team members can express and critique ideas, promoting a collaborative design process.

4.8 Prototyping and Testing

Prototyping involves the creation of tangible representations of design concepts at varying levels of detail, serving as a tool to explore, communicate, and evaluate ideas (Hanington & Martin, 2021). It represents the culmination of research and ideation efforts, translating abstract concepts into tangible forms.

They can take various forms, ranging from low-fidelity sketches and wireframes to high-fidelity interactive mockups or functional prototypes. Low-fidelity prototypes are employed in iterative design development processes to solicit critical feedback based on realistic scenarios. These prototypes can range from simple props and role-playing sessions to physical or digital prototypes with functional elements tested in authentic field situations (Suri, 2003).

On the other hand, high-fidelity prototypes are more refined, often mirroring the final product's appearance and, in some cases, basic functionality. These prototypes are valuable in later-phase evaluation testing, enabling feedback on aesthetics, form, interaction, and usability from clients and users.

Experience prototyping is a dynamic method that encourages active participation in the design process by engaging users with prototype systems, services, products, or spaces. While traditional prototyping often involves passive viewing for concept communication and review, experience prototyping goes further, enabling individuals to immerse themselves in a live experience with the design concept (Buchenau & Suri, 2000). Like role-playing and simulation exercises, low-fidelity prototypes or props are used to construct realistic usage scenarios and experiences. This method is advantageous for its affordability and feasibility in situations where real-life experiences are impractical due to risks, logistical challenges, or complexity (Hanington & Martin, 2021).

In this project we have created both low-fidelity and high-fidelity prototypes used to explore the implications of a Pet ID system.

5 DESIGN PROCESS

5.1 Mix of Design Phases

The design process is a continuous mix of research, exploration, iteration, testing, and reflection phases (Figure 6). We began with research to understand the landscape we are working within. Once the big picture was painted, we identified a few specific areas where we could make meaningful contributions. Recognizing that time and resources were limited, we had to prioritize some aspects of this speculative world.

With a clearer focus, we moved into the exploration phase, sketching out ideas, prototyping, and engaging in participative activities. The feedback gathered led to synthesis and reflection.

Due to the nature of the subject, reflection plays a key role in the process. We analyzed what works, what does not, why, how, and what that means. This led us back to further research, more sketches, new prototypes, and new reflections. The iterative cycles ensure that each version of our concept aligns more with the needs and realities we uncover along the way.

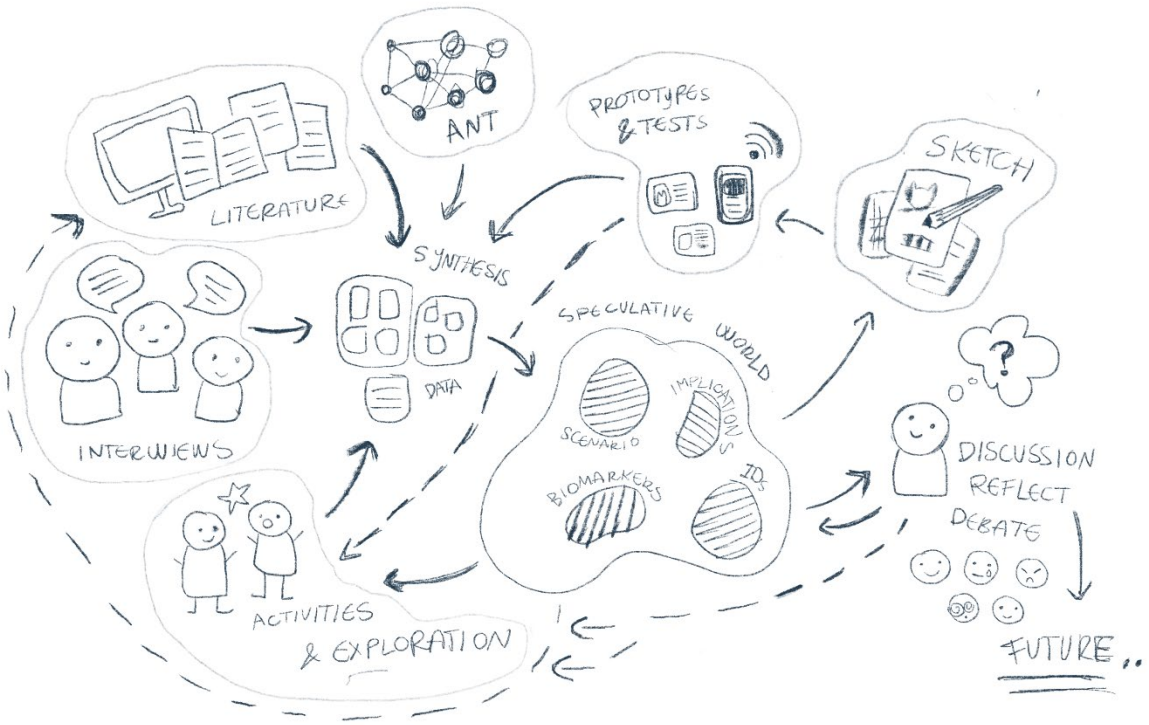


***Figure 6** – Illustration of the design process*

5.2 Reviewing the literature.

The main platforms for the literature review were online search engines and digital libraries (e.g. ACM, ResearchGate). Some of the keywords used in the search include: “animal care”, “animal rights” “biometrics in animals”, “identification in animals”, “animal-computer interaction”, “speculative design”. The outcome of this research can be read in the previous chapters, as well as going forward, supporting this project’s steps.

5.3 Understanding the network with ANT.

5.3.1 The actors

The first step in this ANT analysis (Figure 7) was to identify the current actants involved in the process of animal identification.

1. **Human actors:** Caretakers, Co-Caretakers, Neighbors, Veterinarians, Border Officers, Animal Police Officers, Government Representatives, Lawmakers, Tech Developers, Equipment Makers, Animal Rights Activists, Bus Drivers, Service Operators (store cashiers)
2. **Non-human actors – Animals**: Dogs, Cats
3. **Non-human actors – Technology / Artefacts**: Pet ID Physical, Pet ID Digital, Pet Passport, Government Databases, Microchip, Microchip Scanners, Biometrics, Biomarkers Scanners (nose-print & facial recognition), Smart Phones, Laws & Regulations, Registration Protocols, Scanning Protocols, Animal Rights

5.3.2 Significant interactions

The second step was to translate the network and analyze the interactions within.

1. **Registration Process:** The pet and caretaker interact with the authorities or veterinarians for the registration process. They also interact with the scanner devices, while their biomarkers are getting read.

2. **Data collection:** Authorities or veterinarians record relevant data about the animal, including vaccination records, medical history, and biometric information such as nose prints or facial features into a database.

3. **Scanning Process:** The pet interacts with the scanner devices while the caretaker interacts with the Pet ID / Passport and the other people responsible.

4. **Laws and guidelines:** Authorities in charge of creating laws and regulations, based on research. Pet caretakers and pets

5. **Updating and maintenance:** Pet caretakers are responsible for keeping their pets' ID information up to date, including updating vaccination records and reporting changes in *ownership* or contact information.

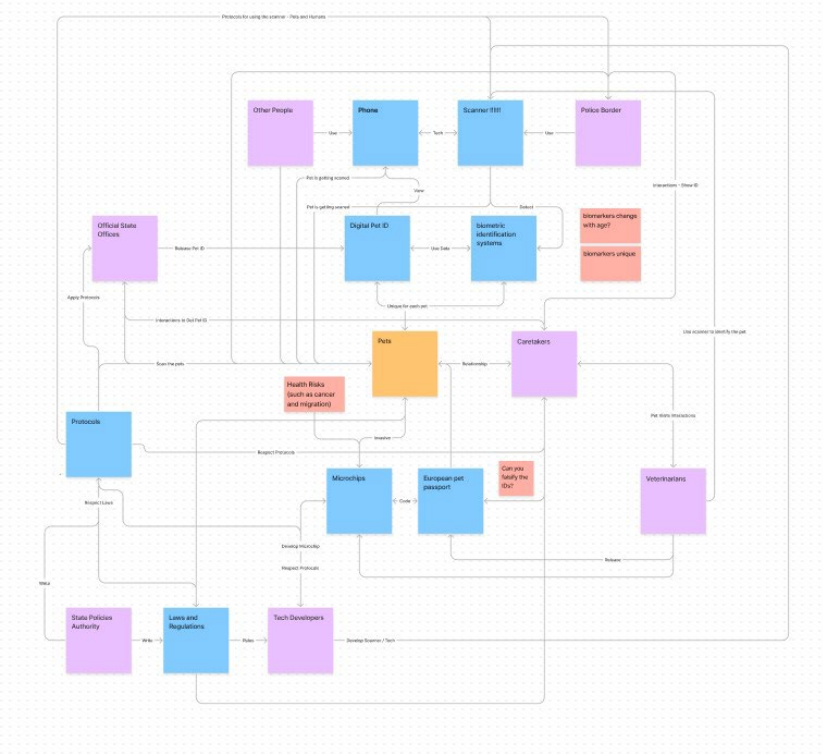


***Figure 7** – Screenshot of ANT analysis*

5.4 User Interviews

Several interviews were conducted with human participants between the ages of 25 and 37, that have experience of caring for animals, either on a personal or professional level. Four participants identified themselves as women and thee as men. The length of each interview

varied, lasting anywhere from 40 minutes up to 90 minutes. Each human participant has received an Informed Consent Form (Appendix 1a), explaining in short about the aim of the project and about the way the data collection and usage takes place. This essential practice for ethical usage of data in research ensures that the individuals can make an informed decision about their participation.

As shown in the figure below (Table 4), the participants and/or their pets are living in different countries in and outside of Europe, so five of the discussions were held in English, and two in Romanian. Depending on availability, the interviews were conducted face-to-face or remote. Having easy access to a multitude of video conferencing software options, we have considered online format to be a good alternative to in-person format. Even though it has some disadvantages, e.g not being able to properly connect with the participants on a body language level, the online format compensates by adding to the participant's comfort and sense of security (Rogers, Sharp & Preece, 2019).

The guiding questions (Appendix 2) were based on the insights gained from the literature review and aimed to dive deeper into understanding each participant's relationship, health care and travelling experiences with their animal. It is important to mention that since the discussions were semi-structured to unstructured on the interview spectrum, each of them took its own shape and they were not consistent across all the participants (Rogers, Sharp & Preece, 2019). This project focuses on dogs and cats, but during the research, there was an opportunity to interview a caretaker who has rabbits (I7), which proved beneficial for our understanding of human-pet relationships.

Interview ID	**Living Country**	**Experience with animals**	**Animals in care**
I1	Sweden	Caretaker	2 dogs
I2	Romania	Caretaker, Veterinarian	2 cats
I3	Romania	Caretaker	1 dog
I4	Indonesia	Caretaker	2 dogs
I5	Sweden	Caretaker	1 dog
I6	Sweden	Caretaker	1 cat
I7	Sweden	Caretaker	2 rabbits

Table 4 *– Interview participants*

5.4.1 Interview Results

By using Affinity Diagrams in processing the data gathered from the user interviews, we have identified a few themes and patterns.

1. **Animals are members of the family.**

   During the interview all participants put accent on the significance of their pets, as they are viewed as individual members of the family. This shared perspective emphasizes the emotional bond between humans and animals.

   While most interviewed caretakers had their well-defined personal motivations for adopting or purchasing their pets, others acknowledged the influence of their romantic partners in the process. One emerging pattern in all the shared stories is how the connections with their pets grew over time. Living closely with their furry companions, even for just a short while, led to a deep emotional bond. Some even mentioned that this bond became even stronger during tough times, like when their romantic relationships ended. Despite the challenges of separation from the partner that influenced the initial decision of getting a pet, they chose to keep custody of their animal. **"*They bring life to an otherwise empty home*"** (I7)

2. **Responsibility falls with the caretaker.**

   Even though they are viewed as important individuals in the family dynamics, animals are still pets and should be treated as such. The caretakers are responsible for the wellbeing of their pets, just as they would be for "***their own kids***" (I4), because of the positions the animals are in: "**they can't go to the store to buy their own food**" (I4).

   In the principles of ACI it is important to consider and design with the specific species in mind. For one, communication takes place in other forms than human-to-human, and we need to find new methods of interaction. It is important to treat each species fairly and accordingly to their needs and it would be a sign of disrespect not to.

3. Managing the caretaking responsibilities.

Co-caretakers of the same pet primarily communicate either verbally or through generic messaging platforms, such as WhatsApp (n.d). They exchange messages, photos, and videos related to their pet, particularly with partners who are currently not living in the same household. Only one interview participant mentioned using other applications, such as Google Calendar (n.d) to help with scheduling in the caretaking process.

Interestingly, none of the participants in the interviews reported an interest in using a dedicated application designed specifically for coordinating and managing pet care responsibilities, considering they are not a necessity for day-to-day interactions and would require too much effort to use.

4. Medical records in physical format.

All interview participants admitted to relying on traditional, physical formats for most of their pets' medical records such as specialized notebooks provided by their veterinarians or simple binders. While not having a digital archive stored on an external server comes with some benefits (e.g in the data security department), having everything on paper also imposes some challenges. Some notable examples that were mentioned during the interviews are: the inconvenience of having to carry or store these records physically; the risk of misplacement or loss brings the potential consequence of losing crucial medical history; and the extra effort put into searching for past specific information.

Taking this further, while some veterinary clinics use in-house digital management systems to process and store patient files, the caretakers often lack direct access to these platforms. This lack of access becomes particularly problematic in case of emergencies or changing clinics. In such instances, they are required to initiate a process of requesting their pet's medical records from the former clinic. This process typically involves receiving the records either as PDF files via email or as printed documents. This results in a fragmented experience for pet caretakers, making it harder to access the medical records and to take decisions about their pet's health.

"***Without a reminder I forget about the yearly vaccine appointments. I don't usually look into the [health] notebook***" (I3)

There is a clear need for a tidier approach for caretakers to store, manage and access their pets' medical records. It points towards a digital solution that would not only alleviate the frustrations expressed, but also enhance the overall efficiency and effectiveness of pet healthcare management system.

5. Physical and verbal reassurance

Ensuring that animals feel comfortable is an important aspect to consider, especially in our project. Without the ability to express themselves verbally, pets often rely on non-verbal cues to convey their emotions, such as body language, vocalizations, or changes in behavior.

When asked about their strategies for comforting their pets during stressful situations, all interview participants emphasized the importance of physical reassurance, such as gentle touching or maintaining proximity. Additionally, most of the caretakers noted the supplementary use of verbal reassurance as an added measure. By remaining attentive to their pets' needs, offering reassurance, and creating a supportive environment, caretakers can help their animals navigate new experiences with greater confidence and ease.

6. Microchips are legally required.

In our interviews, all seven pet caretakers mentioned that their pets are microchipped due to legal requirements. However, only three of them had registered their pets for passports, which were necessary when they relocated to another country. This shows that while microchipping is standard and legally enforced, passports are not a priority unless crossing international borders.

7. Caretakers fear biomarkers are not a reliable identification method.

Multiple caretakers have expressed skepticism about the reliability of biomarkers for use in a Pet ID and feel that the collar tag with the name and caretaker's phone number is the most important identifier in case the pet gets lost.

5.5 Speculative World: The Big Picture

Some of science fiction writer Isaac Asimov's stories revolved around a single technological or societal change and explored its impacts on individuals and society through speculative fiction. In 1942, he introduced the Three Laws of Robotics, and more than 82 years later, these laws continue to shape ethical discussions surrounding human-robot interactions (Anderson, 2017; Encyclopaedia Britannica, 2024). The technological landscape is continuously changing and so is our ethical code. Then we ask ourselves, how might we design interactions that are fair towards other species in a way that makes sense now and in the future?

Embracing a speculative approach in this project is not just about envisioning fantastical or futuristic worlds. It is about using design to imagine, explore and reflect on the present and potential futures.

5.5.1 The big picture: What if Animals Had IDs?

We start this journey by changing just one small aspect of our current society: imagining a world where every dog and cat are identified through a unique Pet ID linked to their biometric data. After going through the initial registration process, each pet is issued a Personal Number, a physical Pet ID Card, and a Digital Profile. The Pet ID contains essential information about their identity and health. It offers them a new level of independence.

While we could go on and on, expanding, building, and exploring this speculative world, we chose to limit our discussion to the following four scenarios and explore how it might impact the relationship between humans and animals: Imagine...

5.5.2 Scenario A: Crossing International Borders

Crossing international borders becomes an easy and smooth experience.

As you and your canine companion approach the border checkpoint, border control officers kindly request both yours and your pet's IDs. Your dog might have aged a couple of years since the photo on the card was taken, but they are clearly recognizable. The officer then uses a handheld scanner to scan your dog's nose. Within seconds, the device confirms your dog's identity. Satisfied with the scan results, the officer returns your Pet ID and nods towards the border gate, signaling your clearance to proceed. With your faithful companion by your side, you continue your journey.

5.5.3 Scenario B: Doggy Day Care Entrance

In this world, even routine activities like dropping your dog off at daycare are made easier and more secure with the Pet ID. At the entrance to the doggy daycare facility, staff members use specialized scanners to verify each dog's identity and access their individual care profiles.

As you arrive at the doggy daycare center to drop off your pet for the day, the staff greets you warmly and directs you to the check-in area. Using a handheld scanner, the staff member scans your dog's nose, instantly confirming their identity and retrieving important information such as dietary restrictions, medical history, and behavioral notes. With a reassuring smile, the staff member assures you that your dog will be well taken care of during their stay.

5.5.4 Scenario C: Transit Across the city by bus

Public transportation systems have adapted to accommodate canine passengers with Pet IDs. Pet caretakers can easily bring their dogs along for the ride.

You and your dog decide to go on a bus ride across the city to visit a nearby park. As you board the bus, you realize you need to buy tickets for your trip. While you scan your bank card in the payment machine, your dog, used to the process of riding the bus, places their muzzle in the Pet ID scanner. The device quickly scans the dog's nose print, and with a beep of approval, it confirms their identity and processes the payment for the pet ticket. Both of you take your seats and enjoy the bus ride to the park.

5.5.4 Scenario D: Lost and Found

When a pet goes missing, the digital system helps facilitate the reunion through coordinated efforts between pet caretakers, authorities, and community members.

One evening, you realize that your cat has gone missing. Frantically searching for them, you quickly mark their status as lost in the Pet ID Digital Platform. Meanwhile, concerned neighbors and community members that have profiles on the platform receive alerts on their smartphones, prompting them to keep an eye out for your lost pet. Within a few hours, a neighbor spots your cat wandering nearby. Using the scanning feature in the digital platform, they scan your cat’s face, and confirm that it is indeed the missing pet. Easily contacting you through the platform, you are reunited with your furry friend.

5.5.5 The small pictures: What we focus on

The big picture is a broad painting, but due to time constraints we chose to focus on a few points of how a Pet ID system might look and work like (Figure 8):

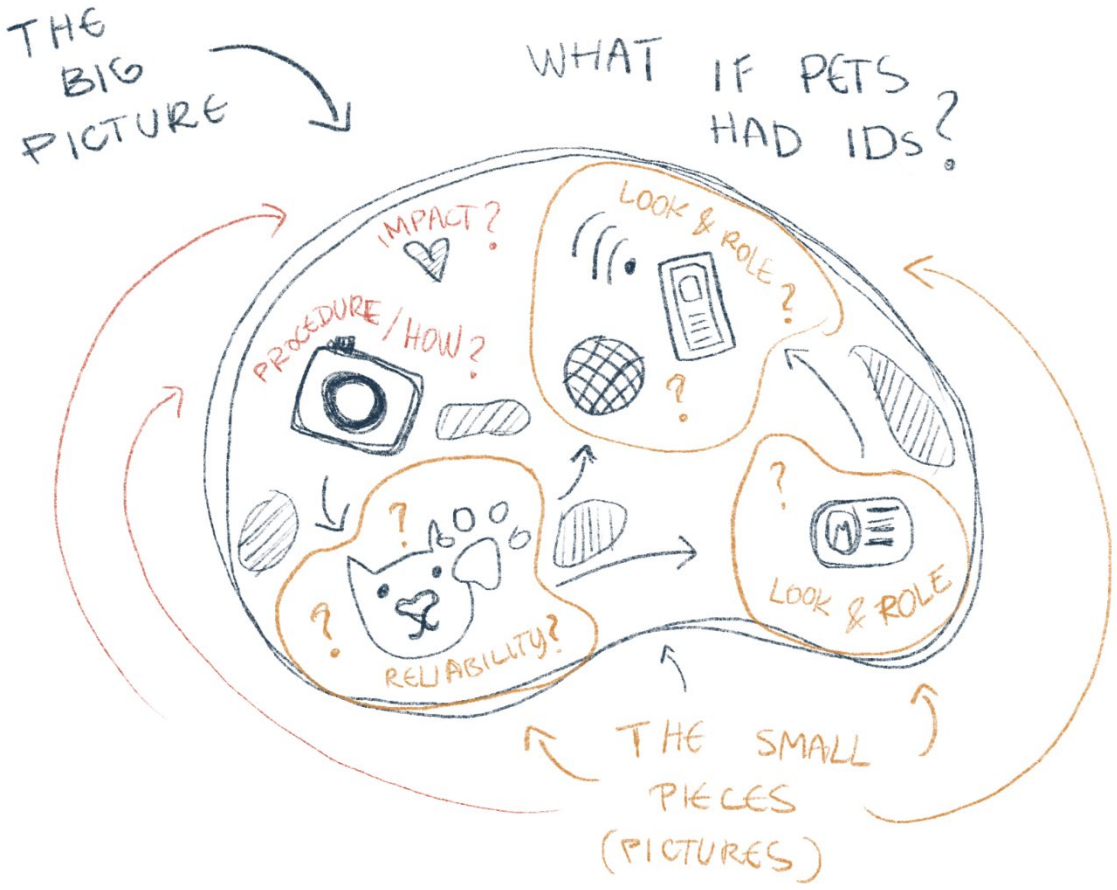


Figure 8 – The small pictures inside the big picture

1. **Physical Pet ID Card:**

   Each pet is issued a physical Pet ID Card. As part of the design process, we created two low-fi prototypes for the physical artefact, described in *Chapter 5.7 Prototype: Pet ID Card* and *Chapter 5.10 Prototype: Pet ID Card 2.0*

2. **Biometric Identification:**

One of the key features of the Pet ID system is biometric identification, which utilizes unique physical characteristics of each pet to establish their identity. For dogs, this may involve scanning and storing their nose prints, which are as unique as human fingerprints, along with a high-resolution photograph for visual verification. Similarly, cats may have their facial print as part of their biometric profile.

To better understand biometrics, we created a workshop activity aimed to experiment with visual biometrics to see how easy it is to recognize an individual solely based on them, explained in Chapter *5.8 Photo Identification Activity*.

3. **Digital Pet ID Platform (Digital Profile)**

Behind each Pet ID Card is a digital profile, accessible through a secure online platform. This digital profile contains information about the pet, including their medical history and emergency contact information for their caretakers.

We created a digital prototype in Figma to illustrate and test these features with users as explained in *Chapter 5.11 Prototype: Pet ID Digital Profile.*

5.6 Sketching

Sketching was an important step in creating the Pet ID system. It provided a space to explore and visualize design options for the Pet ID Card (Figure 9), the Digital Platform and later other concepts that could be a part of the Pet ID system (Figure 21).

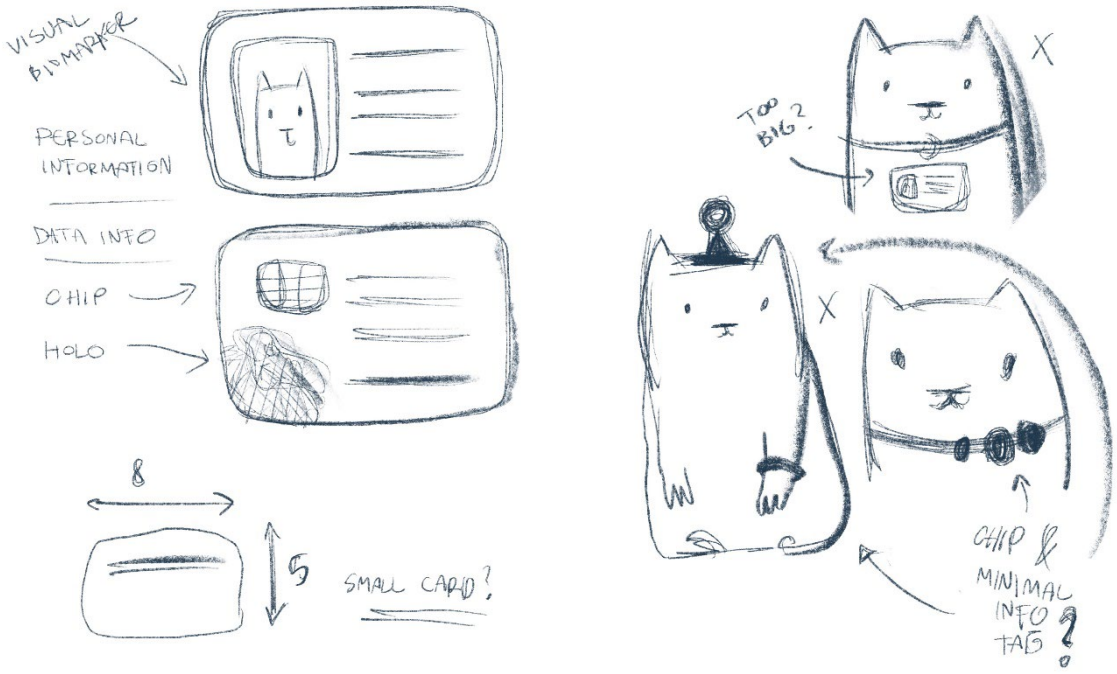


Figure 9 *– Some of the sketches for the Pet ID concept.*

5.7 Prototype: Pet ID Card

Drawing inspiration from the human National ID Cards, explained in Chapter 3.1, the choice to prototype a physical card for the Pet ID was made to connect existing human-centric identification systems with the need for pet identification solutions. As such, the physical card provides a familiar and tangible form that humans easily understand and recognize.

The first prototype for the Pet ID Card (Figure 10a & 10b) represented a tangible manifestation of one part of the project's vision, as well as an artefact used for exploration into visual biometrics exercise, as explained further in chapter 5.8 Photo Identification Activity.

In building this prototype, different materials such as wood planks, plastic and paper sheets of different thicknesses and cardboard were tested. Materials used include cardboard for the thickness and firm feel; thin transparent plastic sheet and tape for protecting the layers of printed paper and for mimicking the glossy plastic-like feel; double-sided tape for putting the front and back sides together and contributing to the pocket compartmentalization. The decision to utilize these materials allowed for rapid iteration and experimentation, enabling us to refine the design iteratively based on user feedback and practical considerations.

The physical Pet ID Card prototype featured two zones for personalization, by including designated pockets for inserting a pet's photograph and minimal personal information (pet's name, pet's birthday, and caretaker's name). This modular design element facilitated easy customization, allowing pet caretakers to personalize the card while testing its usability and practicality in real-world scenarios.



Figure 10a & Figure 10b *– Front and back sides of the prototype: Pet ID Card 1.0.*

During the photo activity, the first complete iteration of the Pet ID Card prototype was put to the test (Figure 11). By engaging participants in this hands-on activity, we were able to observe how the prototype functioned in practice, identify any usability issues or areas for improvement, and gather valuable feedback to inform future iterations of the design.



Figure 11 *–Dog C and their modular Pet ID Card prototype.*

5.8 Activity: Photo Identification

While experimenting with animal biometrics, we looked deeper into understanding the process of distinguishing individuals only by visual biometrics. Starting with the assumption that distinguishing individual animals only through a photo of their face might not be an accurate method of identification by itself, we tested this with a workshop activity.

The human participants were presented with sets of photos for each present animal participant, consisting of one original and other 16 AI generated derivatives (Appendix 3). The task was to successfully identify the original photo of the pet standing in front of them, just as a border police agent would have to.

Throughout the workshop activity, our focus was on two things: first, to observe how participants engage with the images and make judgments about their authenticity, and second, to facilitate meaningful conversations about the implications of pet identification.

Participants

This experiment was completed with three mixed breed dogs and one cat, which we are going to name: Dog A, Dog B, Dog C and Cat A. Human participants consisted of their caretakers and additional individuals that had not met the animals beforehand. This mix of participants enriched the discussions and insights gathered from the experiment. For the activities involving non-human animals, we have sent an Informed Consent Form (Appendix 1b) to those who are legally responsible for them.

Step I

The first step was collecting passport-style (Appendix 4) photos of participants' pets, which served as the foundational material for the experiment. We either took the photos ourselves (see Figure 12a) or asked the caretaker to provide them in cases where we didn't have direct, in-time access to the animals, or the animals were not yet comfortable enough with us.

Step II

The second step was generating a series of visually similar images of each participating animal, by feeding the original photos into an AI Image Generator software (Leonardo.ai). We tried out different prompts and settings (Appendix 5) to test various degrees of image

qualities and resemblances, resulting in 4 batches of 4 generated images (Figure 12) for each animal. The generated images were portraying animals, resembling the original photos in posture and appearance, each with some slight differences.

Afterwards the photos were printed at 3,5cmx4,5cm size and marked with the subject and batch number. With these sets of images, we aimed to closely mimic the characteristics of the participating pets, setting the stage for an intriguing exploration of human perception and technology's capabilities.



Figure 12 *– Generated Image Batches of Dog A, on a similarity scale.*

Step III

Due to the nature of working with animals, it was necessary to prioritize their wellbeing and comfort, so the workshop was split into three separate events. Each activity took place in a familiar environment for the animals, together with their caretaker and other participants (Figure 13a & 13b).

For this activity we used the modular physical Pet ID Card, prototype as described in chapter *5.7 Prototype: Pet ID Card*. Since the personal information and photo sections were left empty, they could be personalized for each animal. Once the caretakers filled in the information zone, each participant received the personalized Physical Pet ID Card artefact and was tasked to identify one-by-one which of the images depicted the real animal in front of them. Participants carefully examined the images, drawing upon visual cues and familiarity with their pets' appearances to make their selections.



Figure 13a & Figure 13b *–Dog A and Dog B activities took place at their home.*

5.8.1 Activity Results

Starting with the assumption that the caretakers would not encounter any difficulties in identifying the real image of their pets, when asked to identify which of the presented images is the original photo, the results were surprising in some cases.

Two out of three caretakers participating in this activity had trouble differentiating some of the *Batch 3* fakes from the original photo (Appendix 6 - tables A, B & D), while the other caretaker correctly guessed every time (Appendix 6 – table C). It is worth mentioning that the caretaker that has successfully identified all the fake images has photography training, and therefore more experience with analyzing images.

Batch 1, *Batch 2,* and *Batch 4*, that had the least similarities with the original photo, were spotted to be fakes in all instances by the caretakers, and in most instances by the other participants.

Following the identification task, participants delved into a facilitated discussion, sharing their experiences, challenges, and observations.

1. **Taking the photos proved to be a difficult task.**

All the participants reported that taking a photo in the passport specific style proved to be a difficult task. When working with animals it is important to acknowledge that without specific training, getting an animal to sit still for long enough to snap the photo might not be that easy. Even while trying to take photos ourselves, we have encountered difficulties and needed to use props and distractions such as toys and food (Figure 14a, 14b & 14c). We consider that this matter is worth pursuing further in future investigations: How might we improve the photo registration process?



Figure 14a, Figure 14b & Figure 14c *– Photo session for Cat A took place in the pet's home, in the presence of their caretaker and the author.*

2. **Personalization of the card expressed an extra level of care.**

The modular design of the prototype card allowed for personalization, adding an extra layer of care that reflected a deeper level of consideration for pets and their caretakers. Participants mentioned that having the real information next to the photos felt more like a real situation and helped them become immersed. However, the gender was not modular, which caused a problem in one situation because the gender on the card was incorrect. This highlights the importance of paying attention to details, even in prototypes.

3. **Photo is not enough, set of identifiers.**

Relying only on visual biometrics for pet identification may prove insufficient, particularly when individuals are unfamiliar with the pet. Factors like lighting conditions, angle of observation, and the pet's behavior can complicate accurate identification. Therefore, incorporating additional forms of identification, such as nose prints, seems essential.

5.9 Activity: Roleplaying

We have prepared four speculative scenarios for roleplaying (Appendix 7), placed in a speculative alternative world where the focus was on *What if Animals had IDs?*. The participants enacted these scenarios and reflected on the looks of and implications a Pet ID system would have on individuals.

Scenario A – Lost and Found: the lost pet is found by a neighbor that scans the pet's biomarkers and gains access to the Pet ID digital profile and the caretaker's contact.

Scenario B – Border Control: while crossing international borders the pet's identity needs to be verified by authorities.

Scenario C – Veterinary Emergency: the pet, in need of emergency medical care, needs to give the veterinarian access to their medical history records.

Scenario D – Doggy Day Care: access to locations and services, such as a day care program, is based on the pet's identity and, as an extra thought exercise, a *social score* concept.

1. **What information should the ID have.**

   While exploring the role-playing scenarios, the discussion touched on what information the Pet ID should contain. Personal information after which the pet can be identified or labelled, such as name, species, breed, gender, birth date and personal number, were unanimously considered essential.

2. **The concept of social score.**

   Additionally, we explored the concept of a social score, much like an episode of Netflix's dystopian futuristic series Black Mirror (Wright, 2016) or, more recent and real, China's social credit system (Canales & Mok, 2022). We enacted a scenario where the ID might grant access to locations or privileges based on the pet's behavior history.

   The feelings about this were mixed. It could improve interactions between humans and between humans and pets. Potentially between pets as well, if we can design a way for them to understand the system. At the same time, the concept of social scores opens new opportunities for discrimination. We can say it even encourages it by promoting or punishing whatever is considered *good* or *bad behavior* at the time.

3. **Declaring the pet lost.**

The issue of pets getting lost was also debated, recognizing the anxiety and distress it causes for both the pet and the caretakers. This led to brainstorming ways to address this issue more effectively. One idea that emerged was to incorporate visual cues to indicate when a pet is reported lost and needs help.

5.10 Prototype: Pet ID Card 2.0

After analyzing the results from the early testing sessions, we iterated on the prototype of the Pet ID card, adjusting visual details and adding more features. The 2.0 prototype is not modular anymore; it represents a partially fictive canine individual born in Romania and registered in Sweden, based on the author's dog (Figure 15).



Figure 15 *– Pet ID Card, Digital Mockup*

1. **Adjusted size**

Recognizing the need for convenience and portability, the Pet ID Card's size was reduced to the size of a standard bank card (8,6 cm by 5,4 cm) for an easier fit inside the caretaker's wallets.

2. **Bilingual and updated personal information**

In addition, the card was changed to be bilingual, in English and the language of the country of origin, like current European passports for humans (Chapter 3.1.2). This feature helps ease the language barriers when traveling abroad.

The card's front side displays the pet's personal information, including name, species, breed, gender, and birth date, alongside a personal number assigned upon registration in the international database.

On the back of the card, there's the microchip linked to the pet's Digital Profile, along with a unique card code for easy tracking. For extra protection, the registration date, expiration date and who issued the card is also visible. As additional pet personal information we also included their registered residential address and caretaker.

3. **Visual Biometrics**

When identifying pets using visual biometrics, using only a static photograph has limitations. A photo can change with lighting and angles and may not accurately capture the pet's unique facial features. Fear, discomfort, or movement during the photo session can also affect the image.

To address this, we recorded a short video of the pet's face, from which we extracted a still frame to print on the Pet ID Card. The motion short provides a more accurate representation of the pet's facial characteristics from various angles. To simulate this in the prototype, we used an AR (Augmented Reality) tool to link the video to the photo.

4. **Changing Pattern**

Pets get lost sometimes, and identification is helpful for reunification with their caretakers. On the front side of the card, along the right margin, we added a pattern that changes color when a pet is declared lost. This pattern shifts from blue to red, providing a visual indicator of the pet's status (Figure 16). This dynamic feature is linked to the pet's digital profile.

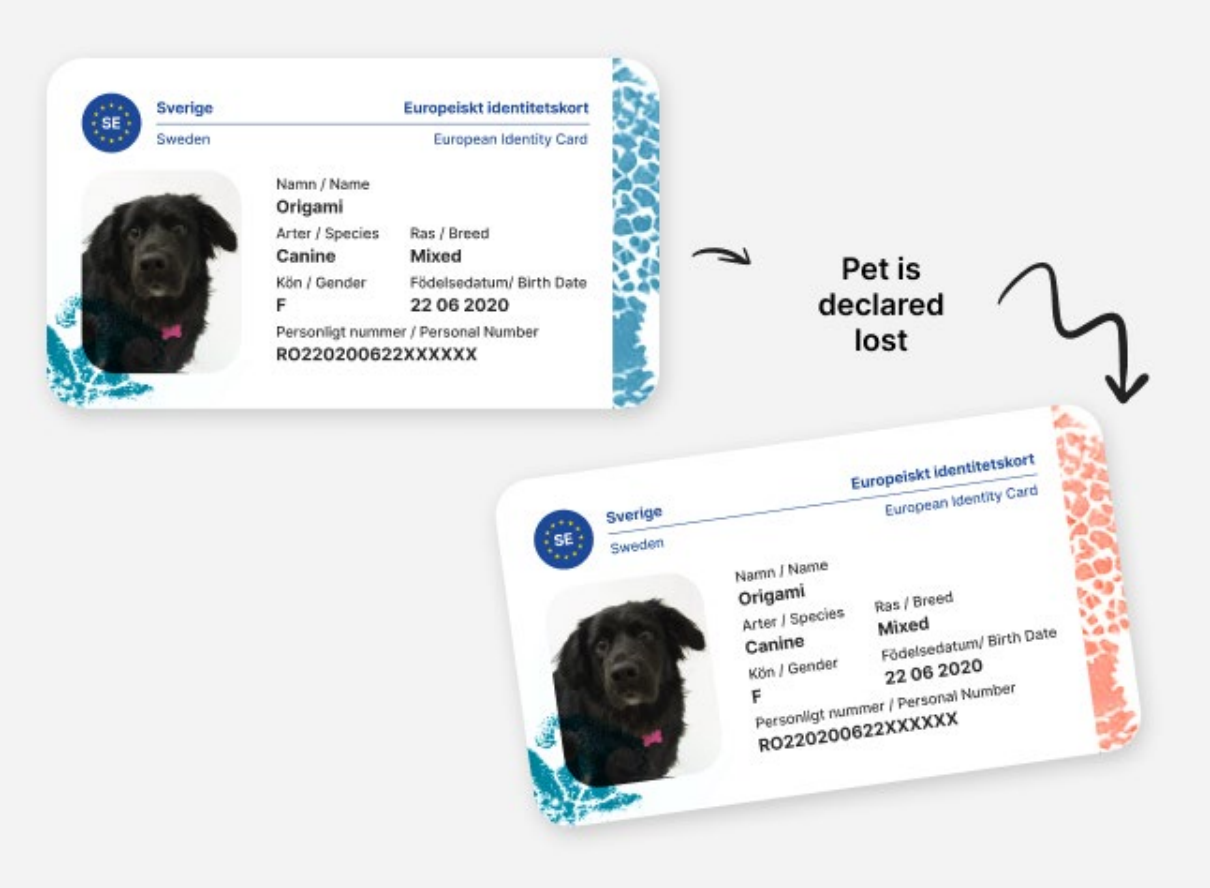


Figure 16 *– Pet ID Card, Digital Mockup*

5.11 Prototype: Digital Profile

Creating the digital prototype for the Pet ID Platform and Digital Profile involved several stages, from concept development to implementing key features.

1. **Feature Definition:** Drawing from the speculative scenarios in *Chapter 5.5*, we defined the key features and components of the Pet ID Digital Profile. This included features such as navigating the pet profile, scanning a biomarker, and declaring a pet as lost.

2. **Figma Screens:** We translated the design concept into high-fidelity screens using Figma, a collaborative design tool. Figma allowed us to create an interactive prototype with visual details and simulations.

This prototype aims to initiate a discussion about the implications of a pet ID by providing an interactive platform for the caretakers to test and warm up to the concept (Figure 17).

! Try out the prototype here !



***Figure 17** – Pet ID Digital Profile Screens*

5.11.1 Log in and Biometric Scanning

Login to the Pet ID Digital Profile is a two-step process. Users enter their pet's unique personal number and password as a primary authentication method (Figure 18a). For extra security, the next step involves biometric confirmation, such as a dog's nose print or a cat's facial features. This ensures that the correct pet's identity is confirmed before granting access. The prototype includes a low-fidelity scan simulation feature (Figure 18b), allowing users to experience the biometric confirmation process.

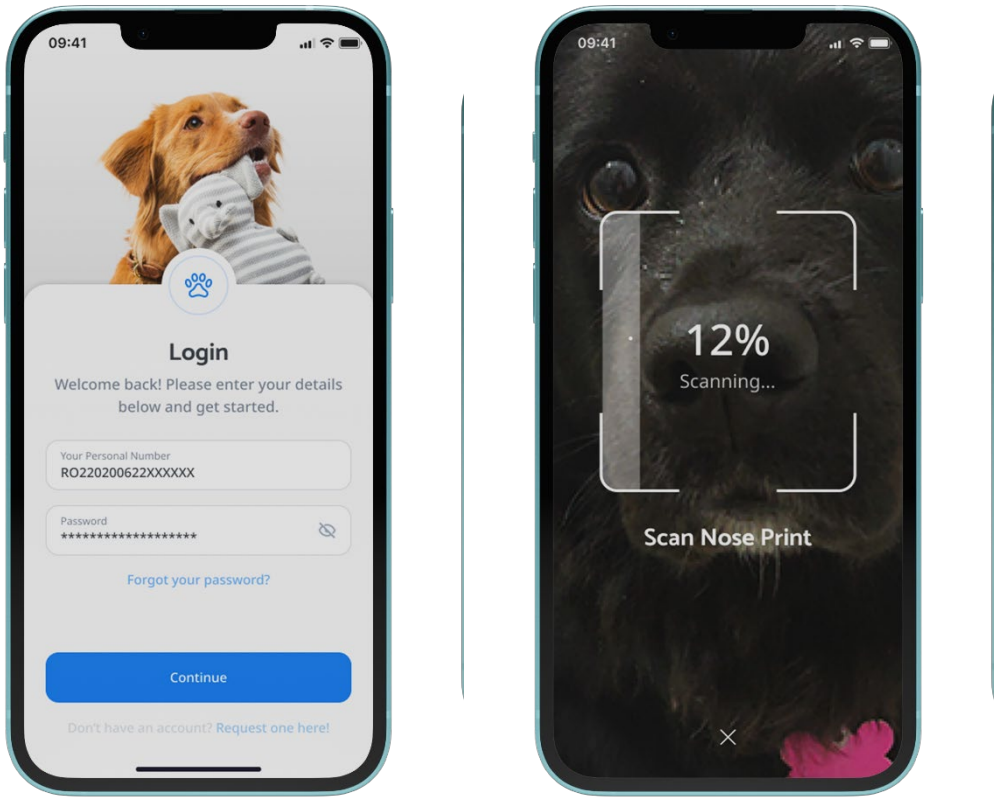


Figure 18a & Figure 18b *– Pet ID Digital Profile screens: Log in & Biomarker Scan*

5.11.2 Pet Digital Profile

The Pet Profile feature is designed to provide a user-friendly interface containing the main aspects of the pet's identification. The flow begins with a dashboard, from where users can access the necessary information and functionalities.

1. The Pet Profile page offers the option to control or preview what information appears on the public profile (Figure 19a).

2. The personal information page allows users to declare their pet lost or to request an update appointment if any information changes. It features two tabs. The first tab displays the registered pet's information, including personal number, name, species, breed, age, gender, general description, distinctive marks, and residence address. The second tab shows the registered biomarkers and allows users to add extra photos.

3. A virtual copy of the physical Pet ID Card, with all the information.

4. The Pet ID also records the pet's health history, including vaccinations, treatments, other interventions, and insurance details (Figure 19b).

5. The emergency contacts section includes the caretaker's contact information so that the right people can be quickly reached in the event of an emergency.

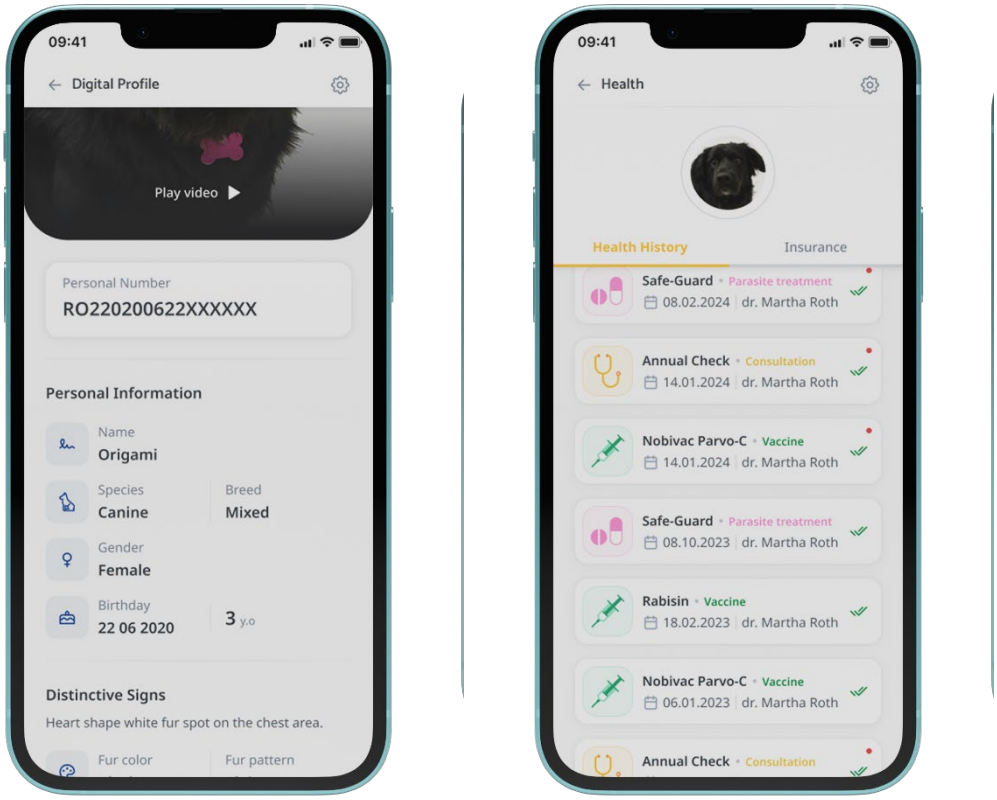


Figure 19a & Figure 19b *– Pet ID Digital Profile screens: View Profile & Health*

5.11.3 Declare pet as lost feature.

When a pet is declared lost (Figure 20a), the system takes action to aid in its recovery. The pet's ID card pattern changes from blue to red and the public profile is updated to show the lost status (Figure 20b). Anyone who accesses the pet's Digital Profile can see that the pet is missing and in case they find them, can contact the caretaker. When the pet is found the status can be updated.

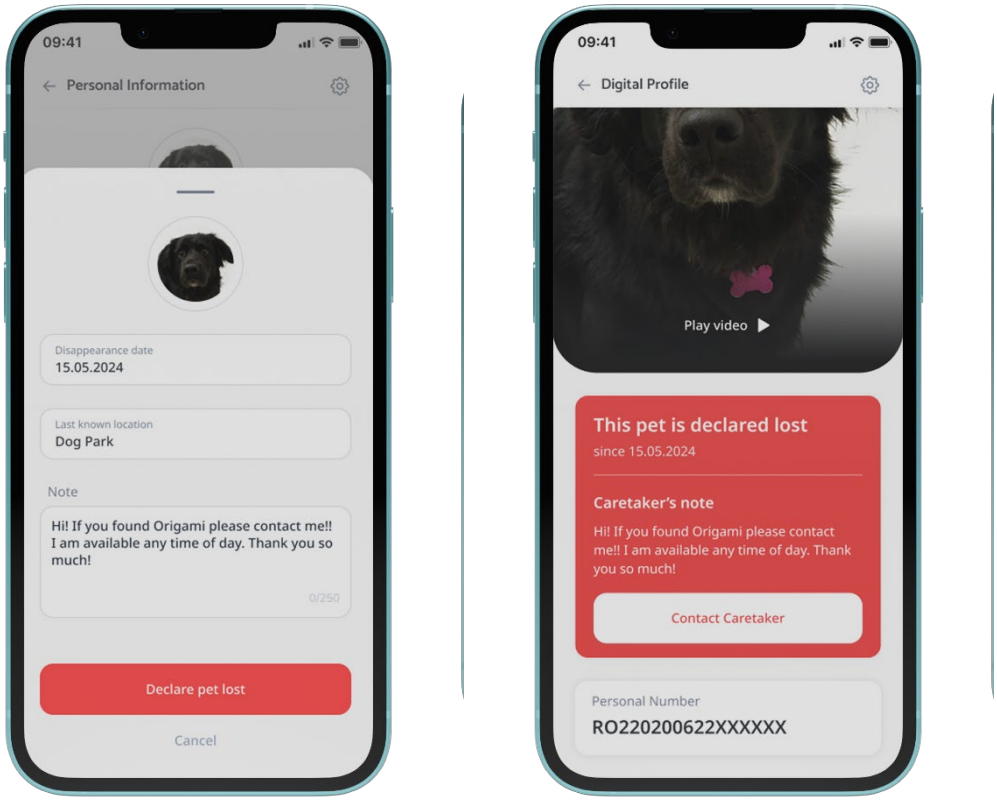


Figure 20a & Figure 20b *– Pet ID Digital Profile screens: Declare pet lost & Public Profile when pet is lost*

5.11.4 Prototype Testing Results

A few testing sessions were conducted with human participants between the ages of 24 and 30, that are caretakers of pets (Table 5).

Testing ID	Living Country	Experience with animals	Animals in care
T1	Sweden	Caretaker	2 dogs
T2	Sweden	Caretaker	1 dog
T3	Romania	Caretaker	2 dogs
T4	Sweden	Caretaker	1 cat

Table 5 *– Pet ID Digital Profile testing participants*

1. **Privacy concerns**

   All of them were generally open to using a digital Pet ID but expressed concerns about data privacy. ***"The only concern I can see right now is about the public information displayed. I think there should be an option to hide the address and contact information unless the caretaker chooses to share it in case of emergency such as the pet getting lost.*** **(T3)**

   "I like that the ID is also in the application, something that would be easy in case you lose the physical one, but at the same time it can be a bit worrying that it is so easy to access the respective data on the ID." **(T2)**

2. **Transparency about biometric data**

   Some participants suggested additional features, including more detailed explanations about the biometrics used. This highlights the need for transparency in how biometric data is handled. **"*I would like the application to explain the difference between nose print and facial print, why there are two options.*" (T2)**

3. **Alerts for emergencies**

   Additionally, the suggestion for an alert feature when a pet is lost pointed to a practical feature that could significantly aid in reuniting lost pets with their caretakers. ***"[...in case of emergency such as the pet getting lost.] I think an alert in that sense could be good too."*** **(T3)**

6 DISCUSSION

6.1 Biometrics in animals

6.1.1 Concerns about biometrics as an identification method

While studies indicate that nose prints are unique to each dog (Choi et al., 2021), are not yet trusted as a reliable form of identification by society. During the research, caretakers have expressed two main concerns about the accuracy of biomarkers and how susceptible they are to change over time.

1. The appearance of a dog's nose may change over time due to unexpected factors such as injury or illness and these changes could potentially alter the nose print.

2. There is a possibility of errors, including false positives (incorrectly identifying a non-matching print as a match) and false negatives (failing to identify a matching print).

While we cannot address these concerns at this time, we recognize that the technology is not perfect, but it is advancing at a rapid pace, especially in this field. A recent study (Guo et al., 2024) is challenging the well-known assumption, *"that no two fingerprints, even from different fingers of the same person, are alike"* while arguing that their AI algorithm found a new way to read human fingerprints, therefore increasing the accuracy.

6.1.2 Draft Step Proposal for future study

Below we mention a draft proposal for a potential future study on the reliability of biometrics if the concept of Pet ID is to be considered for international implementation.

1. One of the first steps is controlling the environments in which biometrics are registered by establishing special registry points, whether at veterinarian clinics or dedicated offices. Standardizing factors such as lighting, camera angle, and image resolution and developing specialized species-specific biomarker scanners (Figure 21) will help obtain clear and consistent data.

2. Establishing an international database and protocols for storing the information is essential. The more countries have access to it, the more quantitative and diverse data we can gather for the study. Starting within the European Union is an accessible goal, especially with a similar human initiative adopted earlier this year, The European Health

Data Space (European Commission, 2024), a data sharing framework for health data in the European Union.

3. Cross-referencing biomarkers at a species level is one of the most important steps in this draft. Comparing the nose prints of all registered dogs tests the reliability of biometrics as a unique identification marker.

4. It's important to update a pet's profile based on their developmental stages (Boehringer Ingelheim, n.d.), such as *puppyhood*/*kittenhood*, *adulthood*, and *seniorhood*. Just like in Romania children receive their first national ID card at the age of 14 (Guvernul României, 2005) and renew it at various birthday milestones, animals would need to renew their Pet ID Cards at various stages in their life. Defining stages like *puppyhood*, *adulthood*, and *seniorhood* for pets would require consideration of species-specific characteristics and developmental milestones. Clear guidelines should be established to determine when this renewal is necessary.

5. Conducting longitudinal studies to track changes in a pet's biomarkers over time offers valuable insights. By monitoring nose prints from puppyhood to adulthood and beyond, we could assess how factors such as growth, injury, and aging impact the consistency of nose prints throughout their life.

6. Finally, presenting yearly results on both the cross-refencing and longitudinal studies shows transparency, reinforcing public trust and confidence in these systems.



Figure 21 *– Sketches for the pet biometrics scanner device.*

6.1.3 The need of a set of biomarkers

Microchips are currently the standard for pet identification due to their convenience. As it is implanted under the skin, they provide a permanent ID that cannot be lost or easily tampered with, thus so have been adopted as a legal requirement. Much like the biomarker concept, a scan of the chip reveals the pet's unique ID number, which links to the caretakers' contact information in a national database.

However, at the end of the day, the microchip is a foreign object inside the animal's body, and it is essential to acknowledge that most of its benefits are for the convenience of humans. As animals may be unable to participate actively and intentionally in the identification process, designers must consider usability, accessibility, but also ethical implications when designing for them.

Given the lack of trust in biomarkers' accuracy, we believe including a set of identifiers makes more sense than having just one. In addition to nose prints and facial recognition, there is space to explore the integration of other non-invasive means of animal identification. In humans, there have been attempts at implementing personal identification systems based on alternative biomarkers such as heartbeats (Kollewe, 2015; Hambling, 2019), as proven they are a reliable identification method (Camara et al., 2022).

Moreover, we encourage keeping the non-invasive traditional identification methods, such as collars and tags, as well. They provide the first line of identifiers, as humans can more quickly read the contact information engraved on a tag, than taking extra steps of scanning the other markers. Having a combination of identifiers adds to the reliability and strength of the identification approach as in the two-factor authentication currently used with digital platforms.

6.2 Let's briefly talk about AI.

6.2.1 AI, as a tool in a designer's kit

Generating a personalized data image set allowed for the flexibility to tailor them to specific characteristics or attributes relevant to the experiment (*Chapter 5.8*), such as appearance and pose. Practical limitations also contributed to the conscious choice of using an AI tool (Leonardo.ai). These include the project's short timeframe, access to numerous animals, photo sets, trained photographers, and access to a relevant animal portrait database. We recognize this experiment as a first steppingstone toward the project's vision, which requires further experimentation with a more hands on approach.

6.2.2 AI, as a part of Pet ID systems

Integrating AI (Artificial Intelligence) technology into Pet ID systems can offer several implications, both in terms of significant benefits and considerations.

On one hand, AI algorithms can analyze large volumes of data more efficiently than manual methods. AI-powered image recognition technology can enhance the accuracy and speed of pet identification by matching visual characteristics and other biomarkers. When reading and comparing biomarkers, AI seems to be the smartest, if not the only, choice.

On the other hand, AI decision-making processes can lack transparency. These algorithms rely on access to large datasets, including sensitive information about pets and their caretakers. This alone raises ethical questions regarding biases, animal treatment, and unintended consequences. For example, it could use pet data to predict future trends and outcomes, such as the likelihood of certain health conditions or behavioral issues emerging in specific pet populations. This potential scenario could enhance society's overall well-being by allowing the implementation of preventive measures. However, it also opens the possibility of unfair or discriminatory treatment towards stakeholders.

6.3 Reflecting on the prototypes.

6.3.1 The Physical Pet ID Card

The concept of the Pet ID card reflects the pet's identity within society. However, this card is a human-centered solution designed for caretakers to carry in their wallets and for other humans to read. Considering that the card's role is to represent and empower the pet in society, shouldn't it be the pet's right and responsibility to carry the card themselves? This raises the question: Is the interaction here intended for animals or humans? Are we imposing a human design onto pets?

In this speculative world we have created an identification system for animals which mirrors that of humans. The pet's identity is designed to function within human systems and, in its current form, can only be understood by humans. If animals can learn human skills like driving (BBC News, 2012), they should be capable of using a Pet ID adapted to their abilities. Thus, if we consider pets carrying and manipulating the IDs, we need to rethink the design.

Dogs naturally use their sense of smell to recognize and identify each other. Therefore, designing a Pet ID that utilizes olfactory cues would be more aligned with their natural capabilities and canine behavior (Kirman et al., 2017). The Pet ID could be a small, lightweight device attached to the dog's tail, containing a unique olfactory signature with the dog's natural scent markers. Much like Dog Internet's Dog CAPTCHA (Kirman et al., 2017) scanning the Pet ID would mean to engage in smelling the device positioned on the back side of another individual. In fact, caretakers can also recognize their dog's scent to a limited extent (Wells & Hepper, 2000; Pribylova et al., 2021), so this design could also make sense for humans too. There is power in *the absurd* (Kirman et al., 2020) and our *absurd proposition* aims to highlight the questions asked above: *Who are we designing for?* and *Are we actually imposing a human design onto pets by suggesting an animal identification system*?

6.3.2 The Pet ID Digital Platform

The Pet ID Digital Profile is an online space containing personal information about the animals and granting them certain access and benefits. But a pet cannot take out their smartphone device, open an app, log in with their email address and type in their password. In this speculative world we created, they are still dependent on their caretakers to manage the profiles for them. In their current form, we recognize that these digital profiles are primarily designed for human-use.

However, there is potential in exploring designs specifically created for pets as direct users (Kirman et al., 2017), including the Pet ID Digital Profile. These pet-centric digital profiles could incorporate sensory elements or interactive features created for animals' capabilities and preferences. For example, tactile interfaces, audio or olfative cues could allow pets to directly interact with their digital profiles. By using their nose or paw to touch specific areas they would receive feedback in a manner that aligns with their sensory abilities.

When designing with and for animals it is important to acknowledge that inter-species technology design has a lot of challenges. Probably the biggest challenge for ACI designers is determining how well a design process can address the needs of animals as active users rather than passive subjects or *usees*. (Lawson et al., 2016)

6.4 Societal implications

In theory, implementing pet identification systems will benefit pets and pet caretakers, with both immediate advantages and long-term implications that strengthen their relationships.

In the short term, having a reliable identification system provides pet caretakers with a sense of security and peace of mind. Knowing that their pets are easily identifiable reduces the stress and anxiety associated with the possibility of them getting lost or separated. Moreover, efficient identification systems facilitate fast reunification in the unfortunate event of a pet becoming lost. This means that caretakers are more likely to be reunited with their pets quickly, minimizing the emotional distress and potential trauma associated with extended periods of separation for both humans and animals.

In the long run, implementing pet identification systems could lead to a cultural shift in how society views pets and open new markets and opportunities for technological advances. Pets would be embraced as community members with recognized identities, empowering animals in legal contexts and strengthening the bond between humans and animals.

6.4.1 New job opportunities

As with introducing new technologies and systems, implementing an international Pet ID system, as discussed in this project, has the potential to open a market of new job opportunities, for example:

1. Pet ID Registration Officers, dealing with the registration process of pets, from taking photos and videos to collecting the nose prints and the rest of the personal information.
2. Pet ID Inspector Officers, tasked with verifying the pet's identity, through the pet IDs. This could be an extension to already existing jobs, such as border police officers, but it would require learning additional skills.
3. Pet ID database administrators, responsible for securely storing and updating pet information, troubleshooting technical issues, and ensuring data integrity and privacy.
4. Pet ID technology developers, in charge of creating the scanning devices and other technologies revolving around this.
5. Pet ID technology maintenance workers, making sure the devices are working properly and in charge of repairing them when needed.

The new jobs would require new skills. Drawing a parallel to practices such as vent sexing (Otsuka et al., 2016), which involves the skills of determining the sex of day-old chicks, highlights the complexity of such tasks and the need for training and expertise (Horsey, 2002). As briefly mentioned in *chapter 5.8.1*, while presenting the Photo Identification Activity results, the caretaker that had formal training in photography had the most success in identifying the original photos of the pet. Just like *chicken sexers* train to develop their abilities, individuals interacting with Pet IDs would require preparation not only to accurately register, identify, and verify animals' identities, but also working with the systems.

6.5 More rights, more responsibilities?

With the evolution of animal rights comes an important question: Do animals also gain more responsibilities? Traditionally, rights and responsibilities have been linked in human society. For example, citizens have the right to free speech but also the responsibility to use it without causing harm. Translating this concept to animals would mean empowering them to make more autonomous decisions within their capabilities while also being held accountable for their actions.

While animals can exhibit remarkable intelligence and problem-solving skills, their cognitive abilities limit their capacity to understand and take responsibility for their actions. This raises ethical questions about how much responsibility animals can reasonably bear. We see our pets as our children. Pet caretakers commonly refer to themselves as *PAWrents* (combining the words paw and parent) to their *furbabies*. Culturally, the definition of a child and the age at

which they are considered an adult (Wikipedia, 2024, Age of majority) or legally responsible (Wikipedia, 2024, Age of criminal responsibility) vary significantly. In some countries, the minimum age of criminal responsibility does not exist (Defence for Children International, 2015), in other countries, it is ten years old (Brown & Charles, 2021), while in others, it is over 14 years old. If we culturally struggle to agree on what constitutes a child in front of the law, how can we agree on pets? At the end of the day, humans will be the ones ultimately taking responsibility for their pets.

6.6 Care Expressed in Different Ways

In design, especially when developing systems like Pet IDs, the concept of care is essential, considering the needs, wellbeing, and experiences of all users involved.

6.6.1 Who Takes Care of Whom?

Traditionally, the human caretaker is considered the one providing care to their pet and ensuring their wellbeing and safety. This includes offering access to proper nutrition, a safe and comfortable shelter, necessary medical attention, affection, and attention. However, pets also provide care by offering companionship, emotional comfort, and often improving their humans mental and physical health. This mutual care highlights a dynamic relationship where both parties benefit and support each other.

Given this reciprocal nature of the relationship, the term *caretaker* might seem one-sided and insufficient. It implies a hierarchical dynamic where one party solely provides care, which does not fully capture the mutual benefits and emotional exchanges involved. Instead, using terms like *companions* or *care partners* might be more appropriate.

6.6.2 Identified by our relationships.

Identity cards and profiles typically emphasize individual identification. Besides our identifiers, such as age, gender, the color of our eyes, the personal number given by the government, we also identify each other based on our relationships. Humans are social beings, and our identities are often defined in relation to others. We are someone's child, someone's spouse, someone's parent or caretaker.

The designed Pet ID Card has the primary registered caretaker's name on it. So, the Pet ID system is not just about identifying a pet but also about acknowledging the relationship between the pet and its human companion.

Imagine a world where, instead of individual identity cards, we had relationship cards that highlighted our connections with others. For example, a relationship card could indicate someone as a *parent to*, *child to*, *spouse to* or *pet companion to*. This could change the way we shape our daily lives, from interacting with authorities to mundane tasks. For example, picking up a package from the postal office requires individual identification. A relationship card would allow a spouse to pick up a package addressed to their partner, as their relationship and shared responsibilities are officially recognized.

6.6.3 Do we even need pet IDs?

After all, we leave you with a question: Does identification for pets make sense? Humans are often reticent to share their personal information, particularly biometric data. If humans are cautious about their data being out there, is it fair to impose it on animals, especially since they cannot voice objections or comprehend the implications?

7 CONCLUSION

The project began as an exploration of alternative pet identification methods. As we progressed through the design process, our perspectives changed, not just in terms of how we view and treat animals in society but also in the terminology we use to define our relationships and the fundamental need and concept of identification. This journey involved a mix of interconnecting exploration, iteration, and reflection phases, leading us to question not only how pets could be identified more effectively but also the implications.

The exploratory phases used research methods such as literature review, ANT, user interviews and other participatory activities that contributed to a better understanding of the landscape and the networks we are working with.

During the ideation phases, we created a speculative future world with pets having Pet IDs. We fragmented the big picture by creating speculative scenarios and chose specific areas to explore deeper. This also involved sketching and designing two types of prototypes: physical and digital. These artifacts were later used in exploratory activities with participating animals, their caretakers, and other individuals. The first artifact was a modular physical Pet ID Card, which, paired with a series of AI-generated images, played an important role in the visual biomarker exploration activity discussed in *Chapter 5.8 Photo Identification Activity*. Later, a more refined version of the Pet ID Card prototype was designed with additional security features. The second prototype tested the role, look, and feel of a Pet ID Digital Profile through a series of high-fidelity screens designed using Figma.

Despite the limitations of this exploration, including the small number of participants (both human and animal) and the potentially broad scope of the project, we offer a peek into how a world in which animals have IDs might look and how those IDs could be designed based on our current Identification Systems. While we cannot generalize the results due to the project's exploratory nature and limitations, the discoveries and reflections provide a foundation for further animal identification explorations.

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APPENDIX

Appendix 1a – Informed Consent Form – Interviews

Informed Consent Form

Researcher:	Luminita Nertan
E-mail:	lumi.nertan@gmail.com

Project title: Thesis Project 1 – Luminita Nertan
Period: March 27th – June 1st 2024

Presentation
This is a first-year master thesis project by Luminita Nertan, studying Interaction Design Master at Malmö University, Sweden.

Aim
The aim of the thesis project is to explore alternative identification methods of animals that could replace the existing invasive ones and to open a path towards the needed future steps for creating and implementing an international digital Pet ID.

Implementation
The study will conduct 30–60 minutes interviews to understand the relationship between animals and humans. Participants have been selected based on having professional or personal experience in one or more of the fields relevant to the project. (animal care, health care, AI science, biomarkers, EU laws and regulations)

Data recording and storage
The interview will be in online or physical format, and if needed it may be recorded, with the consent of the participant. The collected data will be securely stored on a password-protected computer and will be kept confidential and anonymous. The collected data will be used solely for the purposes of the thesis project.

We follow the GDPR (General Data Protection Regulation) guidelines of the European Union, making sure participation is voluntary, and respecting the participants' privacy. Following these guidelines data is recorded and stored in a secure manner. At the end of the project, the participants may request the deletion of the personal data gathered.

Participant consent
Participation in this interview is entirely voluntary, and participants may withdraw at any time without providing a reason. If participants wish to access the information they have provided, they may do so by contacting the researcher listed above. If they have any questions or concerns, they may freely ask the researcher.

Right of use
The thesis project will be published in the Diva database and other forums required by Malmö University.

By dating and signing below, you indicate that you fully understand the above information and agree to participate in this study on this basis.

Date:	Signature:

Appendix 1b – Informed Consent Form – Workshop Activities

Informed Consent Form – Workshop Activities

Researcher:	Luminita Nertan
E-mail:	lumi.nertan@gmail.com

Project title: Thesis Project 1 – Luminita Nertan
Period: March 27th – June 1st 2024

Presentation
This is a first-year master thesis project by Luminita Nertan, studying Interaction Design Master at Malmö University, Sweden.

Aim
The aim of the thesis project is to explore alternative identification methods of animals that could replace the existing invasive ones and to open a path towards the needed future steps for creating and implementing an international digital Pet ID.

Implementation
The study will conduct 60-90 minutes workshop activities to test the theories, prototypes and scenarios with caretakers and their animals. Participants have been selected based on having the availability and meeting the necessary requirements for the activities. For some of the activities the participants may be required to provide additional data, in form of photos or videos, ahead of time, which may be used as a reference in generating further content and may be included in the written documentation of the project.

Data recording and storage
The activities will be in online or physical format and will be documented with video and/or photographs, with the consent of the participant. The collected data will be securely stored on a password-protected computer and will be kept confidential and anonymous (with the exception of the facial photographs of the animals which may appear in the written documentation of the project). The collected data will be used solely for the purposes of the thesis project.

We follow the GDPR (General Data Protection Regulation) guidelines of the European Union, making sure participation is voluntary, and respecting the participants' privacy. Following these guidelines data is recorded and stored in a secure manner. At the end of the project, the participants may request the deletion of the personal data gathered.

Participant consent
Participation in this interview is entirely voluntary, and participants may withdraw at any time without providing a reason. If participants wish to access the information they have provided, they may do so by contacting the researcher listed above. If they have any questions or concerns, they may freely ask the researcher.

Right of use
The thesis project will be published in the Diva database and other forums required by Malmö University.

By dating and signing below, you indicate that you fully understand the above information and agree to participate in this study on this basis.

Date:	Signature:

Appendix 2 – Interview Guiding Questions

1. Can you tell me about your pet(s)? You can start with basic information (species, breed, age, gender) and expand with whatever you want to share.
2. What motivated you to adopt or acquire your pet(s)?
3. What role does your pet play in your life and your family dynamics?
4. Can you describe your pet's daily routine and activities?

—

1. (IF THERE ARE MORE PEOPLE): How do you communicate and collaborate with other members of your household or family regarding pet care responsibilities?
2. Have you ever used technology or digital tools to assist with pet care, such as pet management apps, smart home devices or wearable devices? If so, can you share your experience with these tools?

—

1. How do you currently manage your pet's health and veterinary care? Do you keep track of vaccinations, medications, or health issues?
2. Have you ever encountered challenges in managing your pet's health records or accessing veterinary care? If so, can you describe these challenges?
3. Have you ever faced emergencies or unexpected situations regarding your pet's health or safety? How did you handle these situations, and were there any lessons learned?

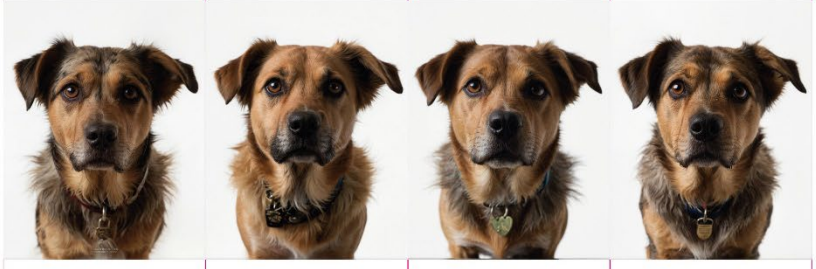
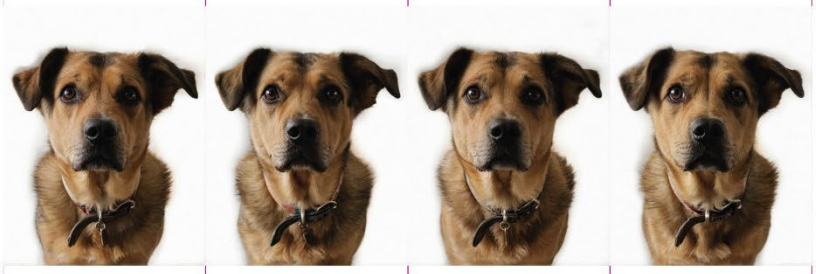
—

1. How do you typically travel with your pet(s)? Do you encounter any difficulties or concerns when traveling with them?
2. How do you ensure the safety and security of your pet(s) when they are outdoors or away from home? How do you calm them down in stressful situations?
3. Are you familiar with the European Pet Passport?
4. Are you familiar with microchipping as a form of pet identification? Have you microchipped your pet(s), and if not, what factors influenced your decision?
5. What are your thoughts on alternative pet identification methods, such as QR code tags or biometric identification? Would you consider using these methods for your pet(s)?

—

1. Can you share any memorable experiences or anecdotes about your pet(s) that highlight the special bond between you and them?
2. Looking ahead, what are your hopes and aspirations for your pet(s) in terms of their health, happiness, and overall well-being?

Appendix 3 – Ai Generated Photos

Dog A Batch 01 (1)	Dog A Batch 01 (2)	Dog A Batch 01 (3)	Dog A Batch 01 (4)
			
Dog A Batch 02 (1)	Dog A Batch 02 (2)	Dog A Batch 02 (3)	Dog A Batch 02 (4)
			
Dog A Batch 03 (1)	Dog A Batch 03 (2)	Dog A Batch 03 (3)	Dog A Batch 03 (4)
			
Dog A Batch 04 (1)	Dog A Batch 04 (2)	Dog A Batch 04 (3)	Dog A Batch 04 (4)
			

Dog A

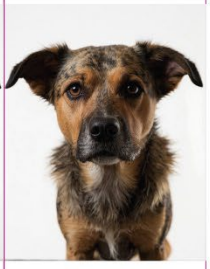
Batch 01

Batch 02

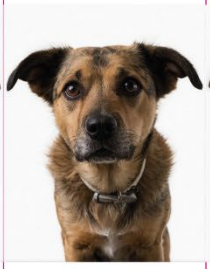
Batch 03

Batch 04

Dog B

Dog B Batch 01 (1)	Dog B Batch 01 (2)	Dog B Batch 01 (3)	Dog B Batch 01 (4)
			

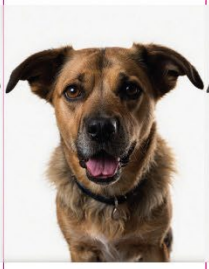
Batch 01

Dog B Batch 02 (1)	Dog B Batch 02 (2)	Dog B Batch 02 (3)	Dog B Batch 02 (4)
			

Batch 02

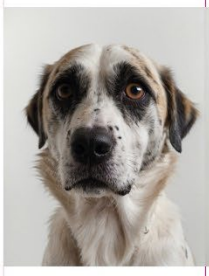
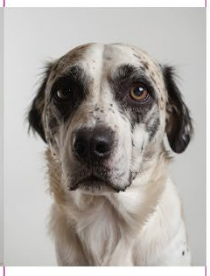
Dog B Batch 03 (1)	Dog B Batch 03 (2)	Dog B Batch 03 (3)	Dog B Batch 03 (4)
			

Batch 03

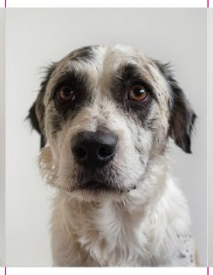
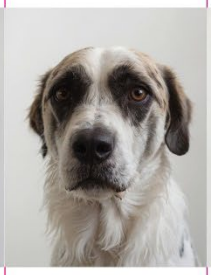
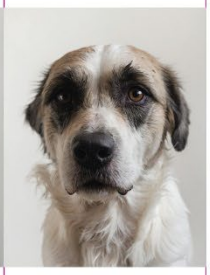
Dog B Batch 04 (1)	Dog B Batch 04 (2)	Dog B Batch 04 (3)	Dog B Batch 04 (4)
			

Batch 04

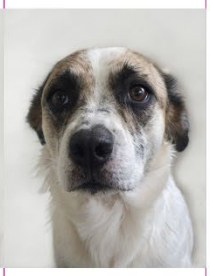
Dog C

Dog C Batch 01 (1)	Dog C Batch 01 (2)	Dog C Batch 01 (3)	Dog C Batch 01 (4)
			

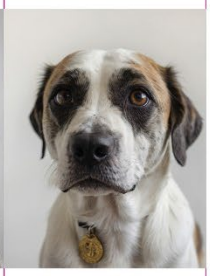
Batch 01

Dog C Batch 02 (1)	Dog C Batch 02 (2)	Dog C Batch 02 (3)	Dog C Batch 02 (4)
			

Batch 02

Dog C Batch 03 (1)	Dog C Batch 03 (2)	Dog C Batch 03 (3)	Dog C Batch 03 (4)
			

Batch 03

Dog C Batch 04 (1)	Dog C Batch 04 (2)	Dog C Batch 04 (3)	Dog C Batch 04 (4)
			

Batch 04

Cat A

Cat A Batch 01 (1)	Cat A Batch 01 (2)	Cat A Batch 01 (3)	Cat A Batch 01 (4)
			

Batch 01

Cat A Batch 02 (1)	Cat A Batch 02 (2)	Cat A Batch 02 (3)	Cat A Batch 02 (4)
			

Batch 02

Cat A Batch 03 (1)	Cat A Batch 03 (2)	Cat A Batch 03 (3)	Cat A Batch 03 (4)
			

Batch 03

Cat A Batch 04 (1)	Cat A Batch 04 (2)	Cat A Batch 04 (3)	Cat A Batch 04 (4)
			

Batch 04

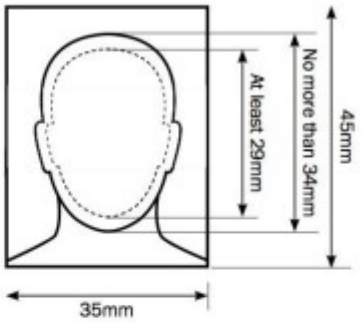
Appendix 4 – Passport Photo Guidelines

according to Swedish authorities

PASSPORT PHOTO DIMENSIONS & GUIDELINES

Paper and sizing

- Photos must measure 45mm by 35mm.
- Photos must be printed on high quality, medium / low gloss or matt photographic paper. heavy weight paper, ie cardboard, is not acceptable.
- Corrugated paper is not acceptable.



Glasses

- We strongly recommend that customers who wear glasses remove them to ensure that an accurate biometric can be obtained. While this may mean that their passport photo does not show them as they usually appear, their photo will still be a good likeness.
- Customers with a visual impairment should also follow this guidance. If they wear dark, tinted or mirrored glasses, they must remove them.

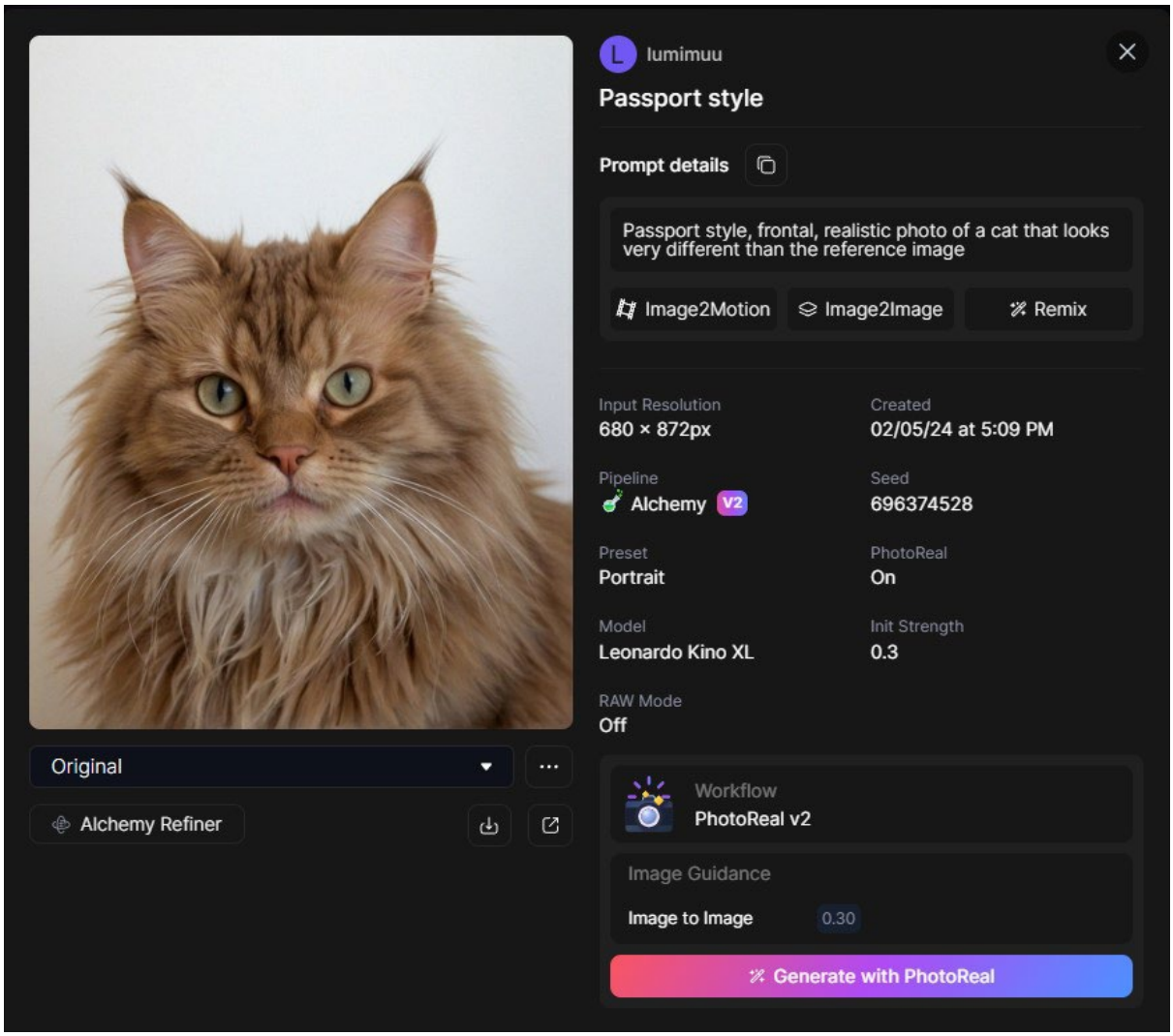
General

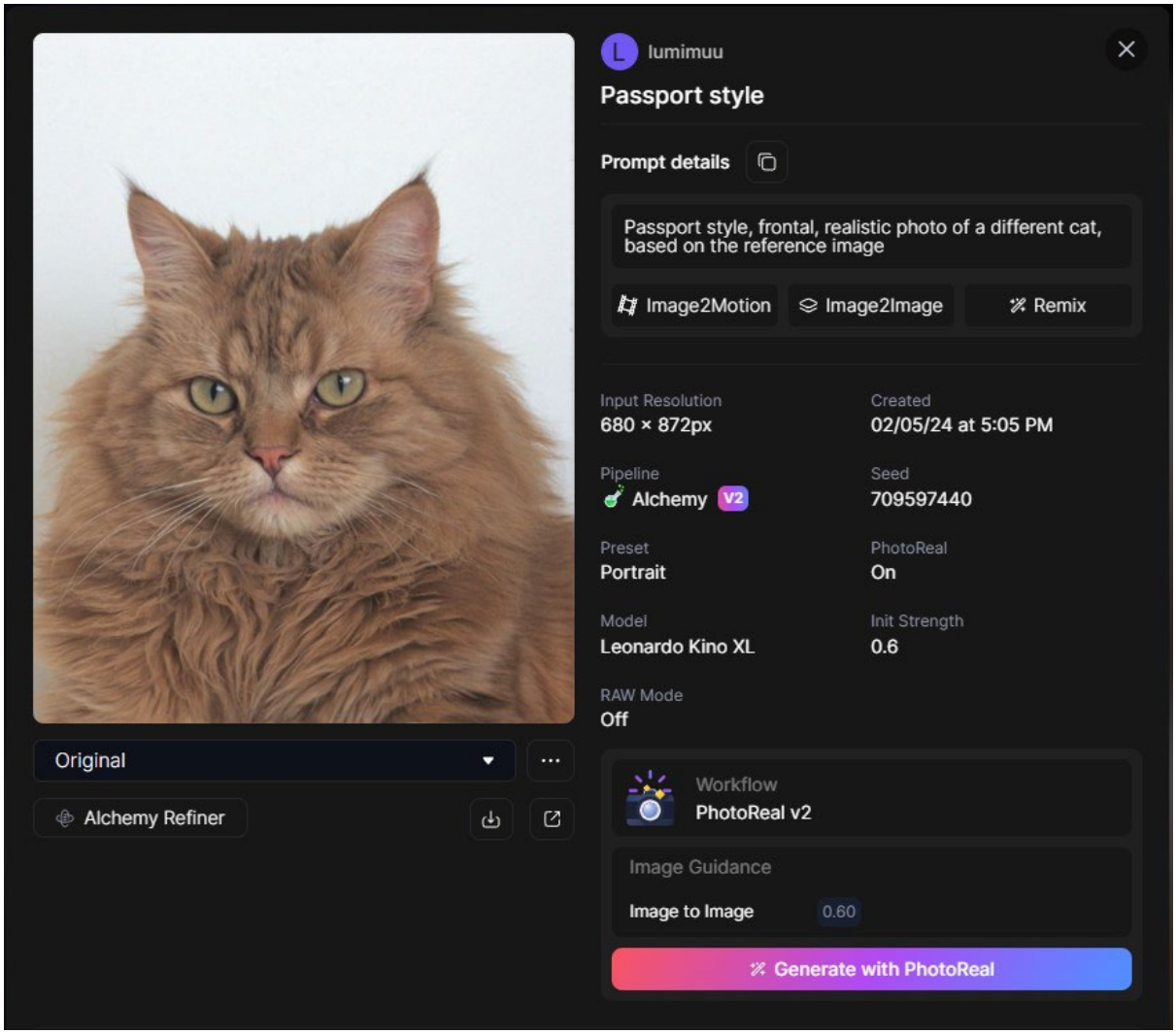
- Photos must be clear, sharp and in focus. They must be a true likeness of the subject.
- Only submit colour photographs; black and white photos are not acceptable. Photos must not be manipulated, for example, by removing spots or softening lines or shadows. Mirror images (that is where the picture is flipped from one side to another, left to right or right to left) cannot be used.
- Where very young children are being supported, you must ensure that no hands are visible. We recommend that infants who are unable to support themselves are laid on a light grey sheet and photographed from above.

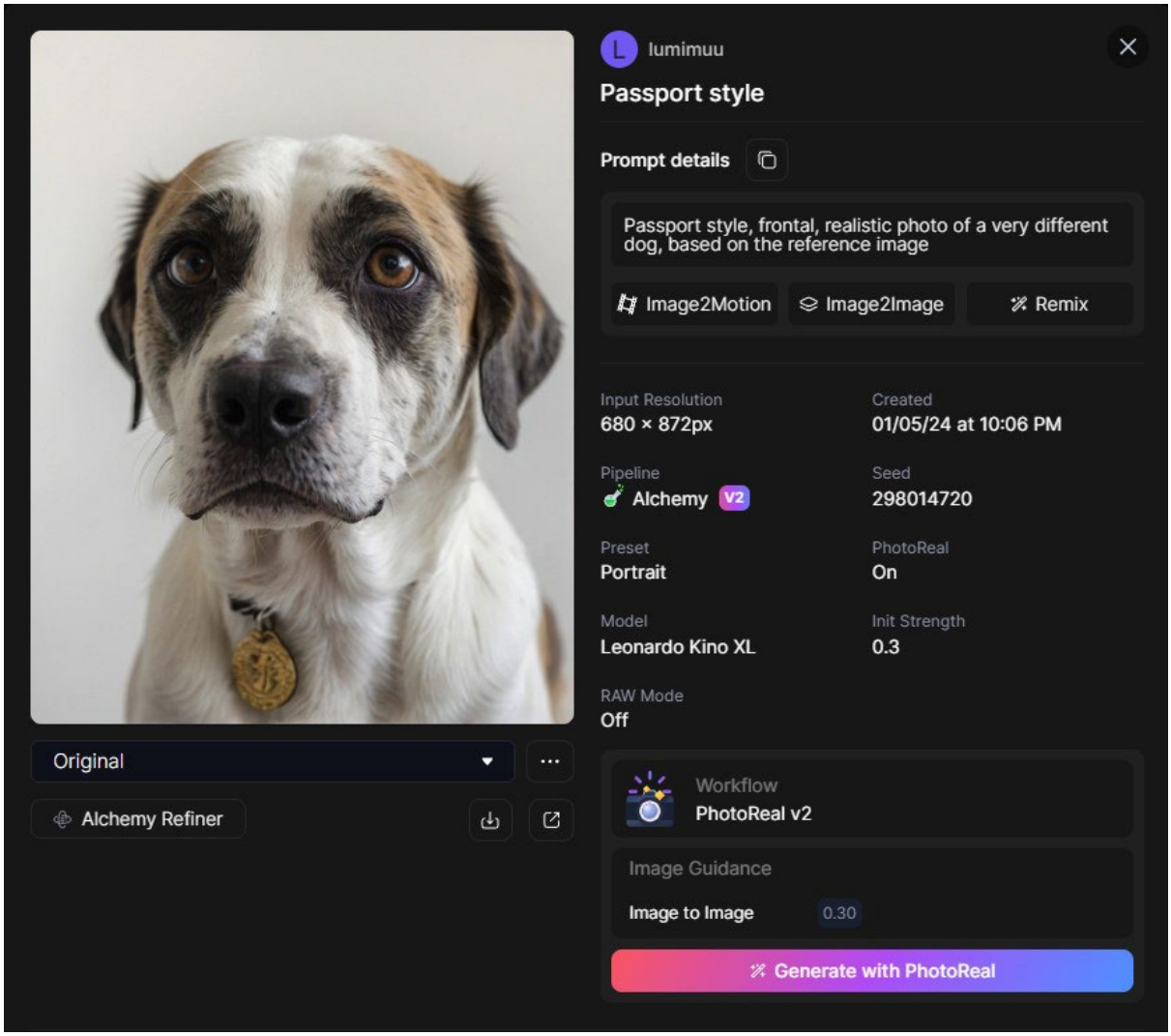
Face

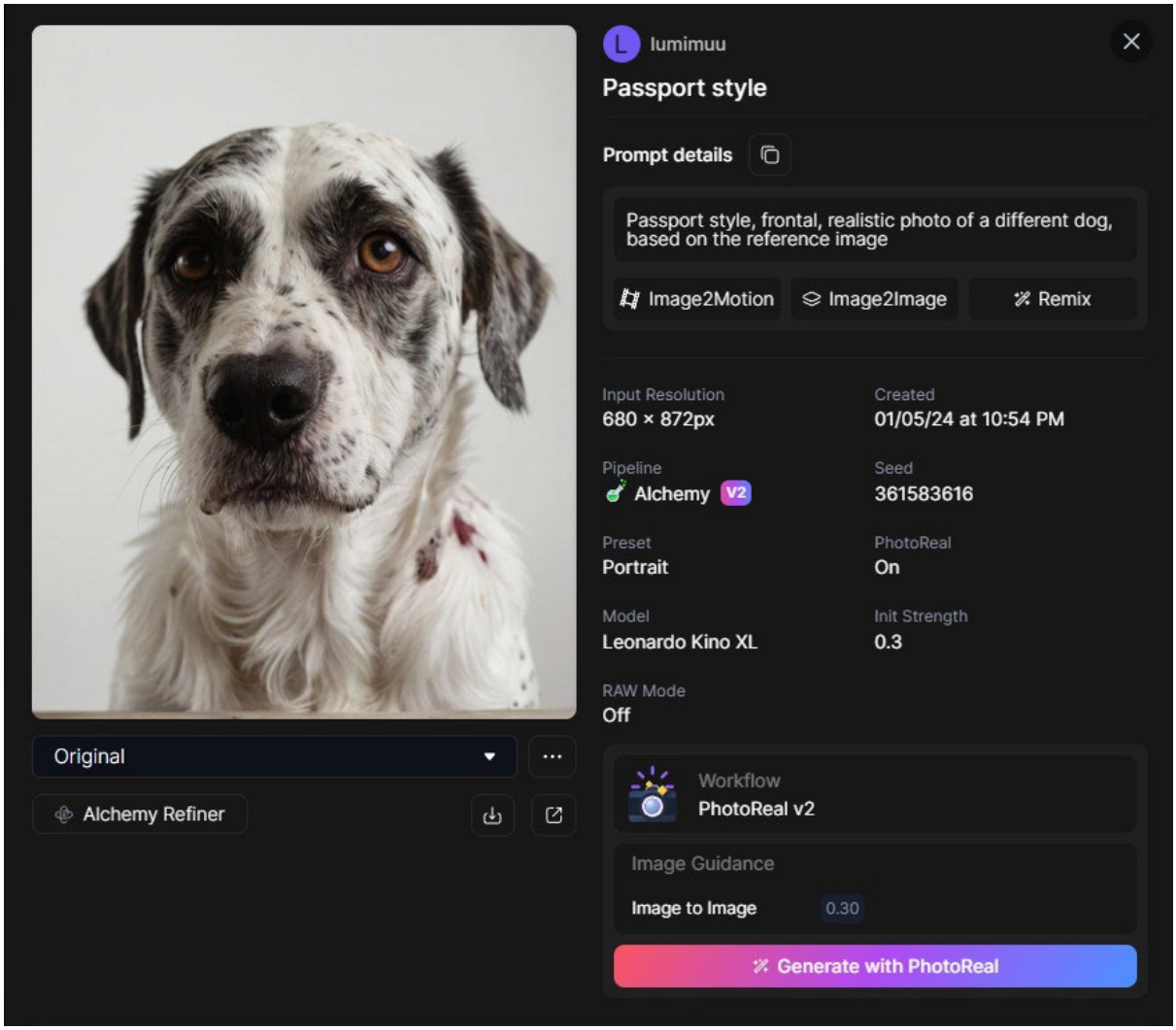
- Eyes must be open and clearly visible, with no flash reflections and no 'red eye'.
- Facial expression must be neutral (neither frowning nor smiling), with the mouth closed.
- Photos must show both edges of the face clearly.
- Photos must show a full front view of face and shoulders, squared to the camera.
- The face and shoulder image must be centred in the photo; do not show the subject looking over one shoulder (portrait style), or with head tilted to one side or backwards or forwards.
- There must be no hair across the eyes.
- Hats or head coverings are not permitted except when worn for religious reasons and only if the full facial features are clearly visible.
- Photos with shadows on the face are unacceptable.
- Photos must reflect/represent natural skin tone.

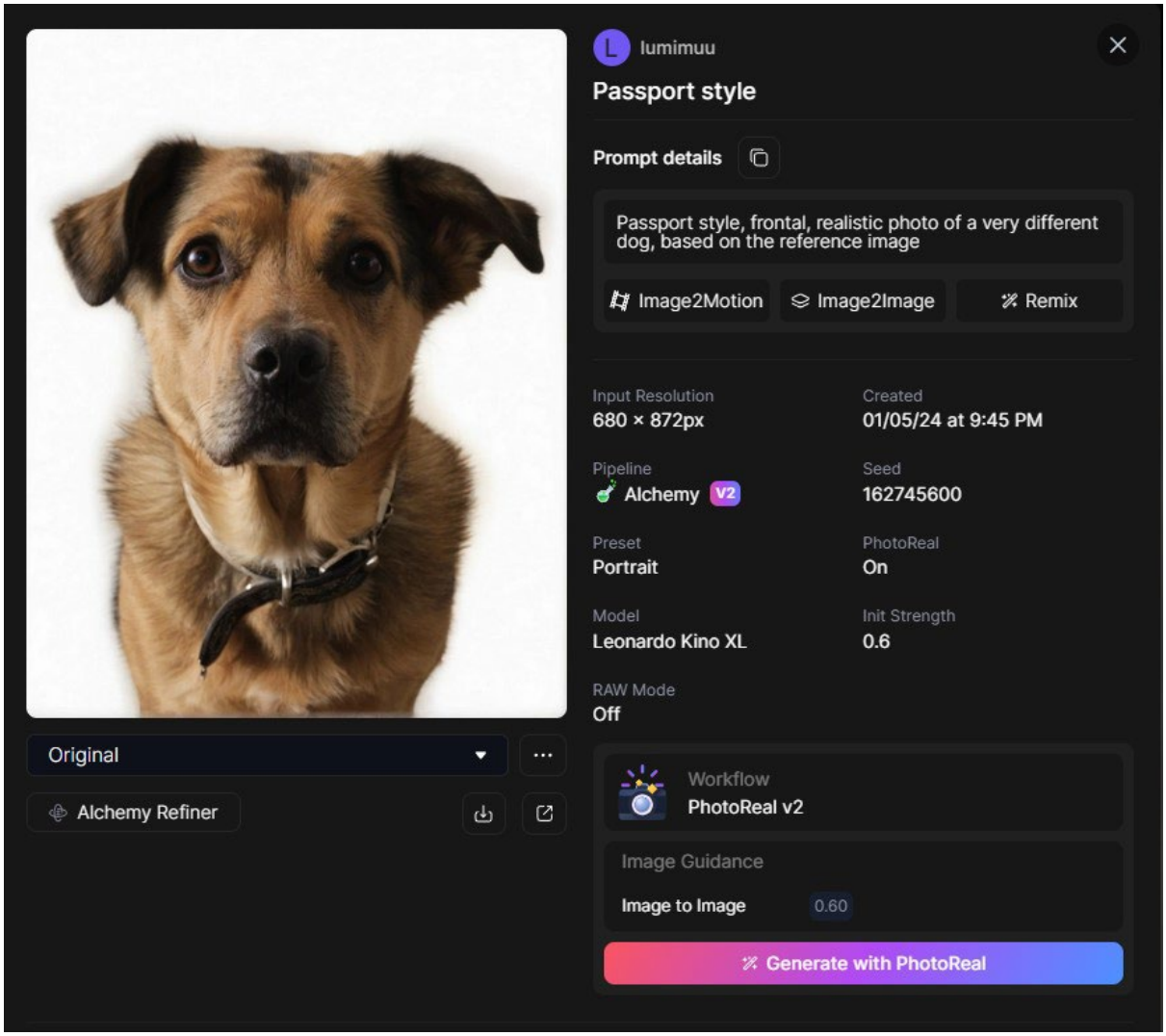
Appendix 5 – Ai Generated Photos Setting




lumimuu
Passport style
Prompt details
Passport style, frontal, realistic photo of a different cat, based on the reference image
Image2Motion
Image2Image
Remix
Input Resolution
680 × 872px
Created
02/05/24 at 5:05 PM
Pipeline
Alchemy
V2
Seed
709597440
Preset
Portrait
PhotoReal
On
Model
Leonardo Kino XL
Init Strength
0.6
RAW Mode
Off
Original
Alchemy Refiner
Workflow
PhotoReal v2
Image Guidance
Image to Image
0.60
Generate with PhotoReal


lumimuu
Passport style
Prompt details
Passport style, frontal, realistic photo of a very different dog, based on the reference image
Image2Motion
Image2Image
Remix
Input Resolution
680 × 872px
Created
01/05/24 at 10:06 PM
Pipeline
Alchemy
V2
Seed
298014720
Preset
Portrait
PhotoReal
On
Model
Leonardo Kino XL
Init Strength
0.3
RAW Mode
Off
Original
Alchemy Refiner
Workflow
PhotoReal v2
Image Guidance
Image to Image
0.30
Generate with PhotoReal


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RAW Mode
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Original
Alchemy Refiner
Workflow
PhotoReal v2
Image Guidance
Image to Image
0.60
Generate with PhotoReal

Appendix 6 – Results from the Photo Identification Activities

Identification Number		**Responses**			
Pet ID	**Image ID**	**Caretaker**		**Participant I**	
Dog A	B1 (2)	No		No	
	B3 (3)	No		No	
	B3 (4)	No		Yes	
	Original	No		Yes	
	B2 (4)	No		No	
	Original	No		Yes	
	B3 (1)	No		Yes	
	Original	Yes		Yes	

Table A *– Dog A activity results*

Identification Number		**Responses**			
Pet ID	**Image ID**	**Caretaker**		**Participant I**	
Dog B	B1 (4)	No		Yes	
	B2 (3)	No		No	
	B2 (2)	No		No	
	B3 (1)	Yes		Yes	
	Original	Yes		Yes	
	B3 (2)	No		Yes	
	B4 (3)	No		No	
	Original	Yes		Yes	

Table B *– Dog B activity results*

Identification Number		Responses	
Pet ID	**Image ID**	**Caretaker**	
Dog C	B1 (4)	No	
	B2 (3)	No	
	B2 (2)	No	
	B3 (1)	No (unsure)	
	Original	Yes	
	B3 (2)	No	
	Original	Yes	
	Original	Yes	

Table C *– Dog C activity results*

Identification Number		Responses					
Pet ID	**Image ID**	**Caretaker**		**Participant I**		**Participant II**	
Cat A	B4 (4)	No		No		No	
	B2 (2)	No		No		No	
	B3 (4)	Yes		Yes		No	
	B3 (2)	No		No		Yes	
	B1 (2)	No		No		Yes	
	Original	Yes		Yes		No	
	B2 (3)	No		No		No	
	Original	Yes		Yes		No	

Table D *– Cat A activity results*

Appendix 7 – Roleplaying Scenarios



Scenario A - Lost & Found

Luna, a curious orange cat, slips out of an open door while her caretaker, Ben, is away. A neighbor finds Luna wandering on the streets, opens her ID and finds Ben's public contact information. Luna and Ben are reunited.

Solution 1:

The neighbor scans Luna's **Pet ID E-Card**

Solution 2:

The neighbor scans Luna's **Biomarker (facial)**

BONUS: Frantic with worry, her caretaker, Ben, logs into the European digital pet ID platform and marks Luna as missing. How Ben change Luna's status in the platform? Does he get notified when someone scans Luna's ID?

To reflect about

What is the ID in this scenario?

How does the ID / Biomarker get scanned?

What information is on the ID?



Scenario B - Border Control

Max, an adventurous german shepherd, accompanies his caretakers, Mary and Bill on a road trip across Europe. At the border checkpoint, immigration officers request proof of Max's rabies vaccination so they can allow the family to continue their journey without any hassle or delay.

Solution 1:

The officer scans Max's **Pet ID E-Card**

Solution 2:

The officer checks Max's **Pet ID Digital version on Mary's phone**

Solution 3:

The officer scans Max's **Biomarker (nose print)**

To reflect about

How does the caretaker present the ID?

How does the ID / Biomarker get scanned?

What information is on the ID?



Scenario C - Vet Emergency

Bella, a playful border collie, injures her paw during a game of fetch in the park. Concerned about Bella's well-being, her caretaker, Elli, rushes her to the nearest veterinary clinic. The veterinarian needs to access Bella's medical history to view her vaccinations, allergies, and previous treatments. With this information readily available, the veterinarian quickly assesses Bella's condition, administers the appropriate treatment, and updates Bella's medical records in real-time on the digital pet ID platform.

Solution 1:

The veterinarian scans Bella's **Pet ID E-Card**

Solution 2:

The veterinarian checks Bella's **Pet ID Digital version on Ellie's phone**

Solution 3:

The veterinarian scans Bella's **Biomarker (nose print)**

BONUS: Bella is scared of the vet. How could Elli and the vet make Bella feel more comfortable?

To reflect about

How does the caretaker present the ID?

How does the ID / Biomarker get scanned?

What information is on the ID?



Scenario D - Doggy Day Care

Rex, an overly active bulldog, eagerly awaits his visit to the local doggy daycare. Before entering the daycare Anna, his caretaker, needs to present Rex's ID so he can be identified as a member.

Solution 1:

Anna presents Rex's **Pet ID E-Card**

Solution 2:

Anna shares Rex's **Pet ID Digital version on her phone**

Solution 3:

Anna presents Rex's **Biomarker (nose print)**

BONUS: Imagine doggy daycare admittance is linked to a Pet Social Score, that is found on the Digital ID. A score under 80 won't allow Rex to enter. What does that mean?

To reflect about

How does the caretaker present the ID?

How does the ID / Biomarker get scanned?

What information is on the ID?